Design, Assembly, and Testing of a Vacuum Hall Thruster

16.522 Final Presentation

Spring 2025

MIT AEROASTRO


PALOMINO
701901
SP

Table of Contents

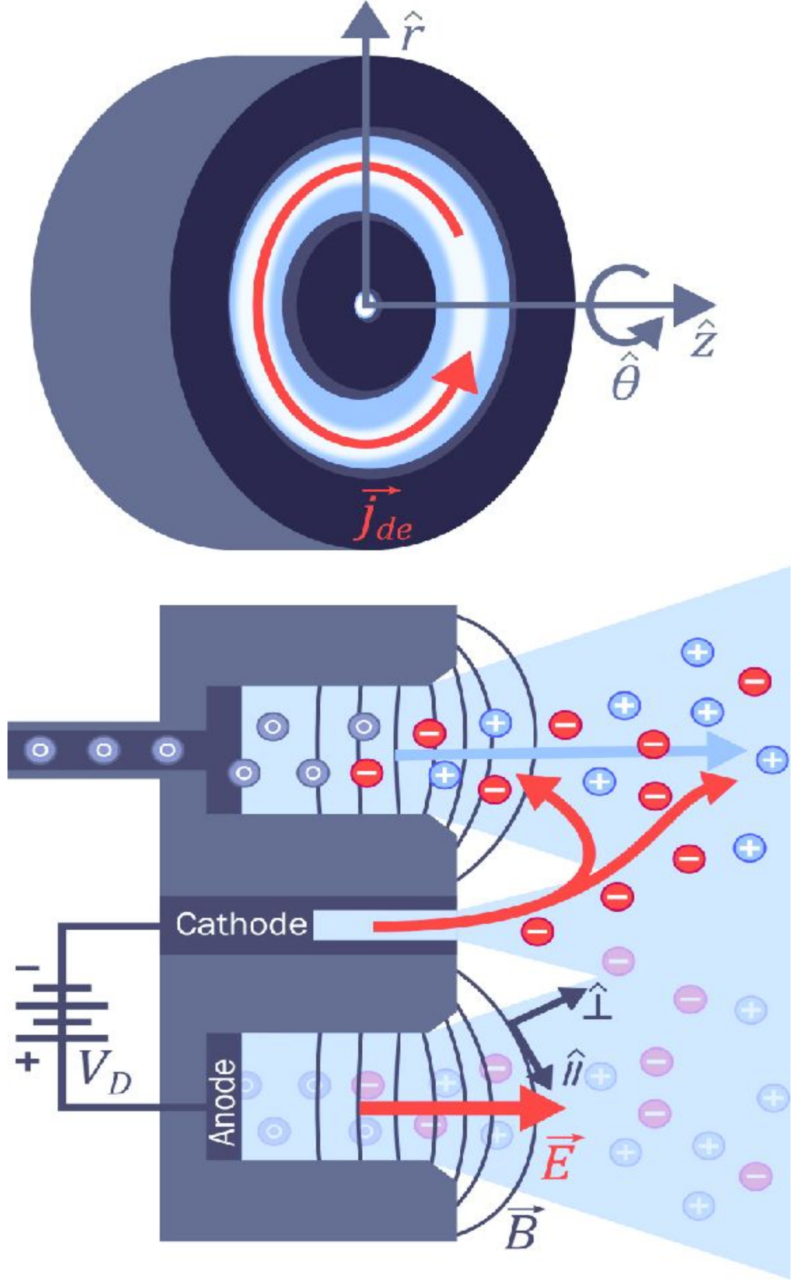
Project Overview

Hall-Effect Thruster Theory & Project Motivation

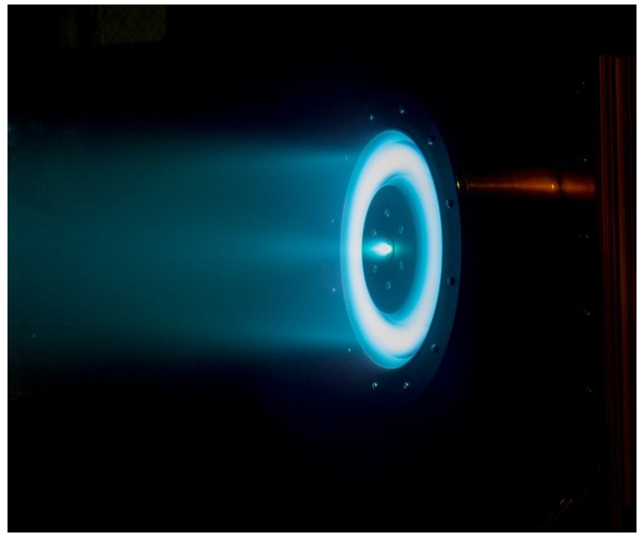
Background

Hall-Effect Thruster Theory

- Electrostatic device (HET)[1]
- Consists of an annular region which permits electron confinement[3]
  - Strong permanent magnet (solenoid)
  - **Field Ionization** through the collision of electron with a neutral atom (Argon)
    - *Larmor Gyration* $r_L = \frac{mv_\perp}{|q|B}$
    - *Ē×B Drift* $\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}$
  - Ions generated from field ionization are accelerated by applied Ē-field
    - Produces thrust to propel spacecraft forward[2,3,4]
  - Ion beam *neutralized* by electrons emitted from the cathode
    - Prevents spacecraft charging



General Hall Thruster Schematic

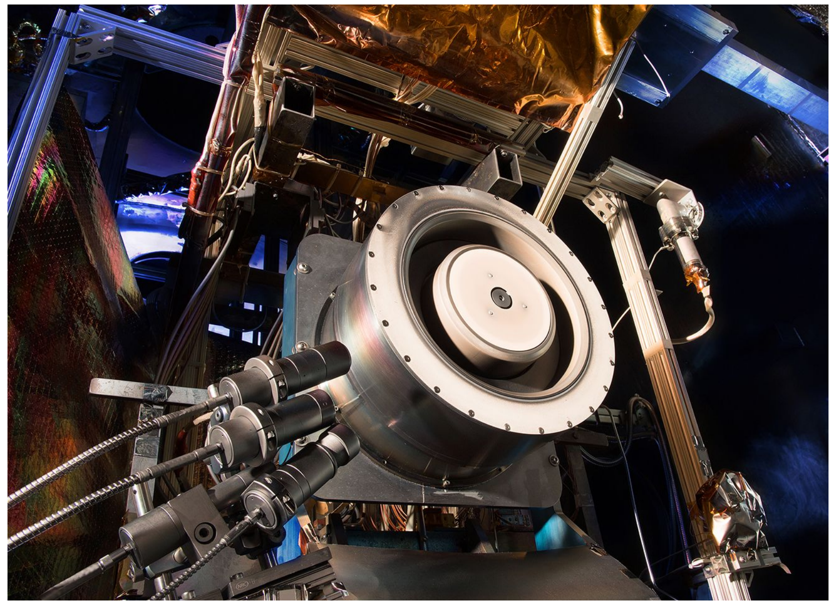


H9 Hall Thruster from University of Michigan

Background

Hall-Effect Thruster Theory

Magnetic Vernier Control & Efficiency Limitations

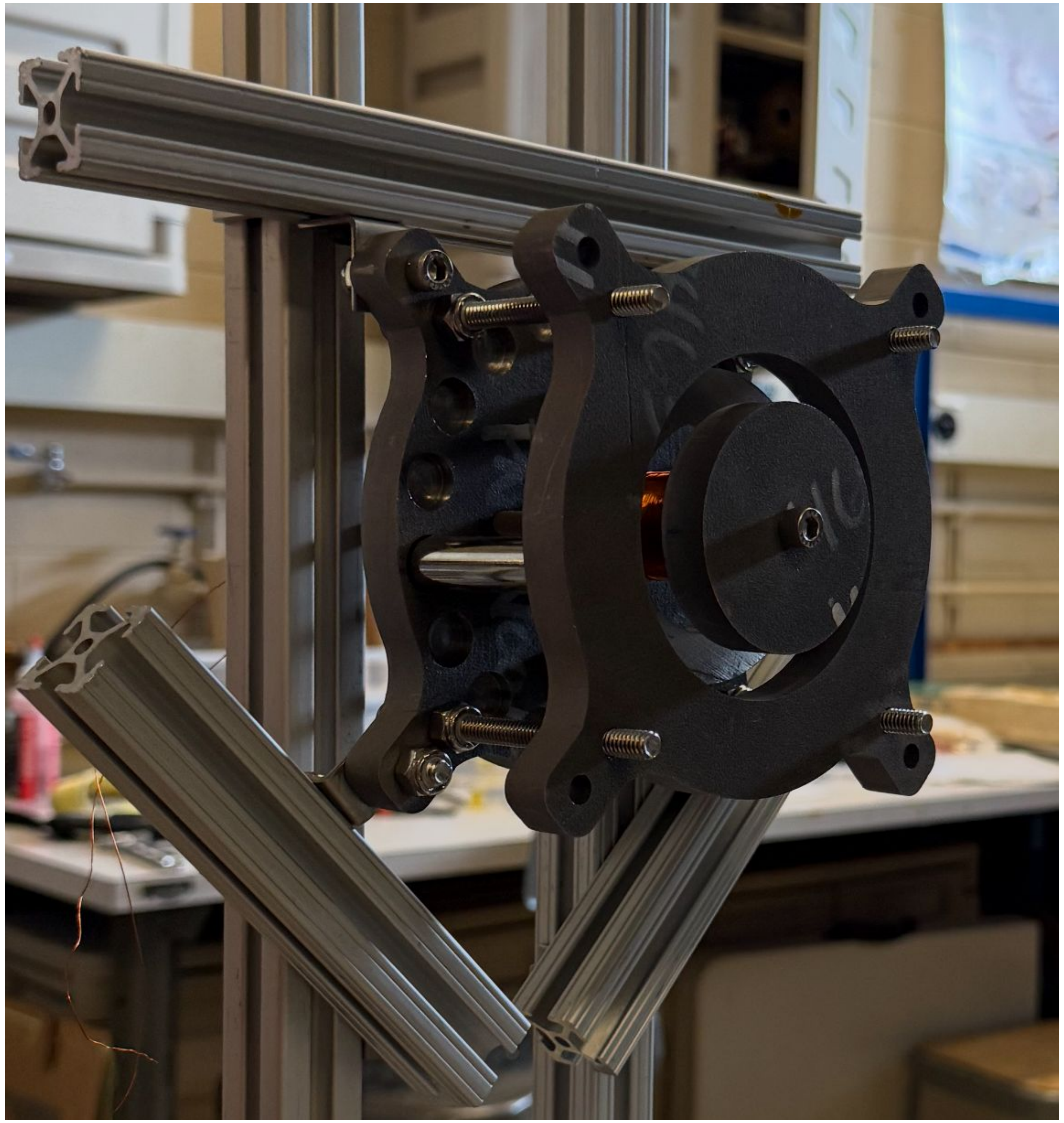


HET from NASA Glenn Research Center

- Magnetic Vernier control permits the modulation of magnetic field geometry and strength
  - Practical for thrust vector control and enhanced maneuverability[4]
  - Modulating B-field can unfortunately lead to plume instabilities

- **HET Efficiency Limitations**
  - 5% of the injected neutral atoms will be ionized
  - Electron drift toward anode due to inelastic collisions (leakage current)
  - Broad distribution of ion velocities being accelerated from ionization chamber
  - Breathing Mode
  - Material erosion due to plasma sputtering[6]

$$\eta = \eta_u \eta_a \eta_E$$



Project Motivation and Objective

Vacuum Hall Thruster with Magnetic Field Vernier

- **Construct Hall Thruster for future LEO missions (orbital maintenance applications)**
  - Argon propellant
  - Power Supply Voltage not to exceed 5 kV
  - Testing at pressures ranging from 1×10^{-3} to 1×10^{-5} torr
- Thruster Dimension Constraints
  - Length < 15 cm
  - Diameter between 8 cm and 15 cm
  - Width between 0.5 cm and 1.5 cm
- Thruster Performance Analysis
  - Ion Beam Current Measurement with resistive load
  - Ion Beam Energy with RPA

Project Objectives

1. Design, build and test a Vacuum Hall Thruster with Magnetic Field Vernier control.
2. Design, build and use a retarding potential analyzer to characterize your VHT.
3. Measure relevant quantities to establish thruster performance.
4. Propose a theoretical framework to predict this performance and justify your design.
5. Postulate and justify the TRL of your VHT design at end-of-effort.

Project Management

Sub-Team 1

Project Organization

Management Policy

Set up weekly Team-Lead meetings:

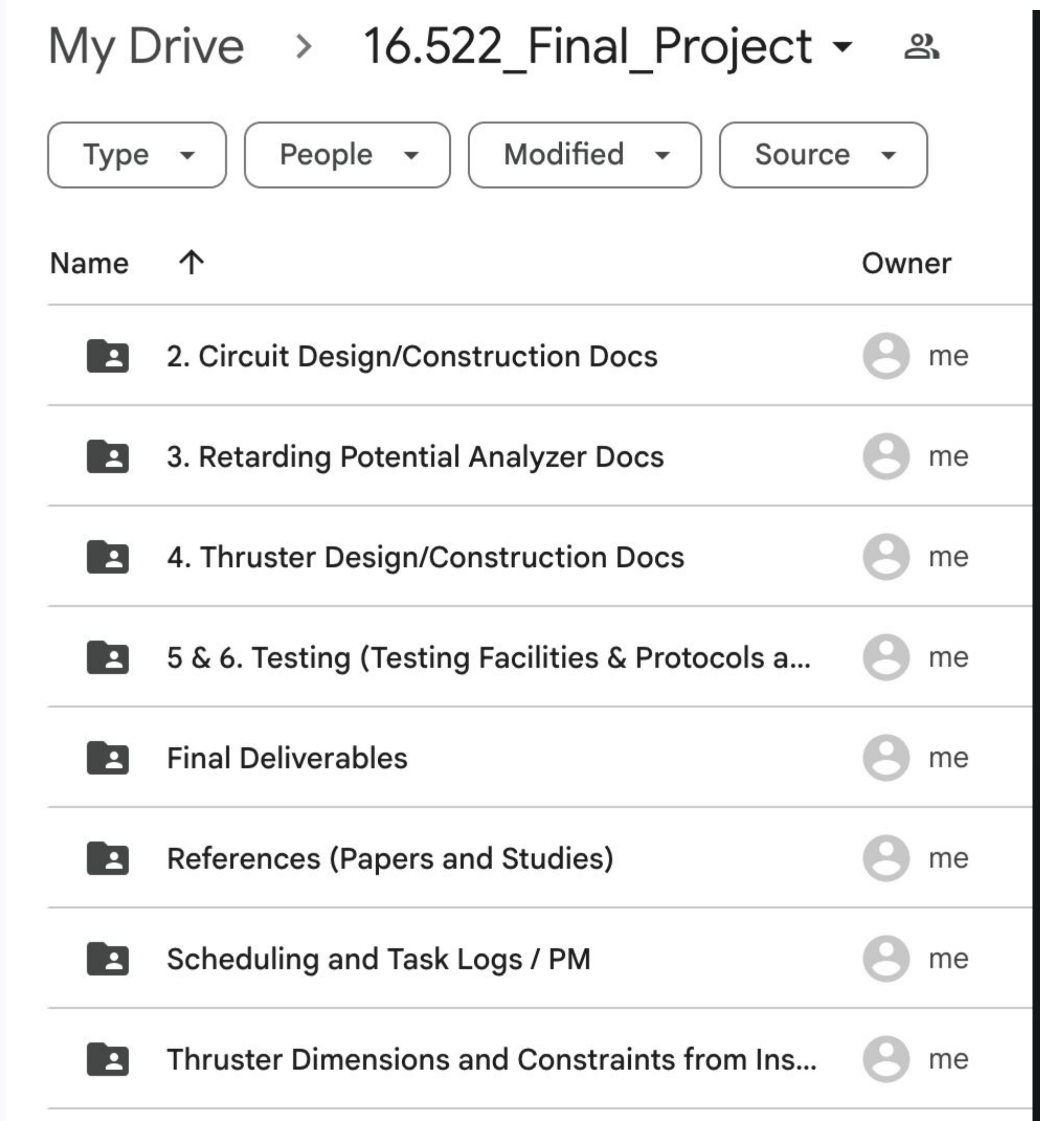
- Established that sub-teams should meet weekly, organize running notes
- Keep project on-track according to timeline
- Discussed interfaces between sub-teams
- Testing facilities and testing analysis teams decided to join for the first part of the project

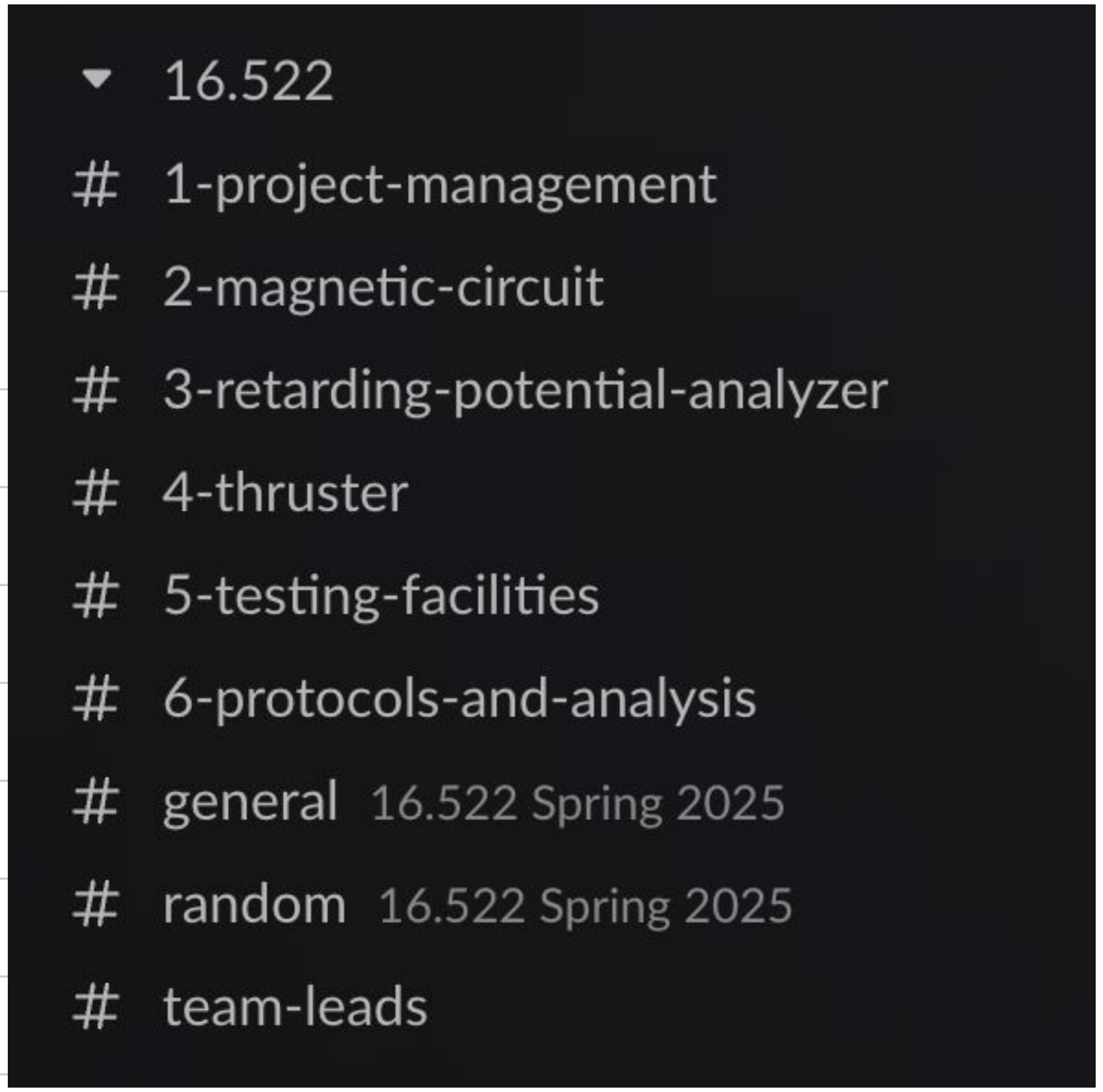
Sub-teams: Project Management, Magnetic Circuit Design, Retarding Potential Analyzer, Thruster, Testing (Facilities & Analysis)

Project Organization

Documentation & Communication

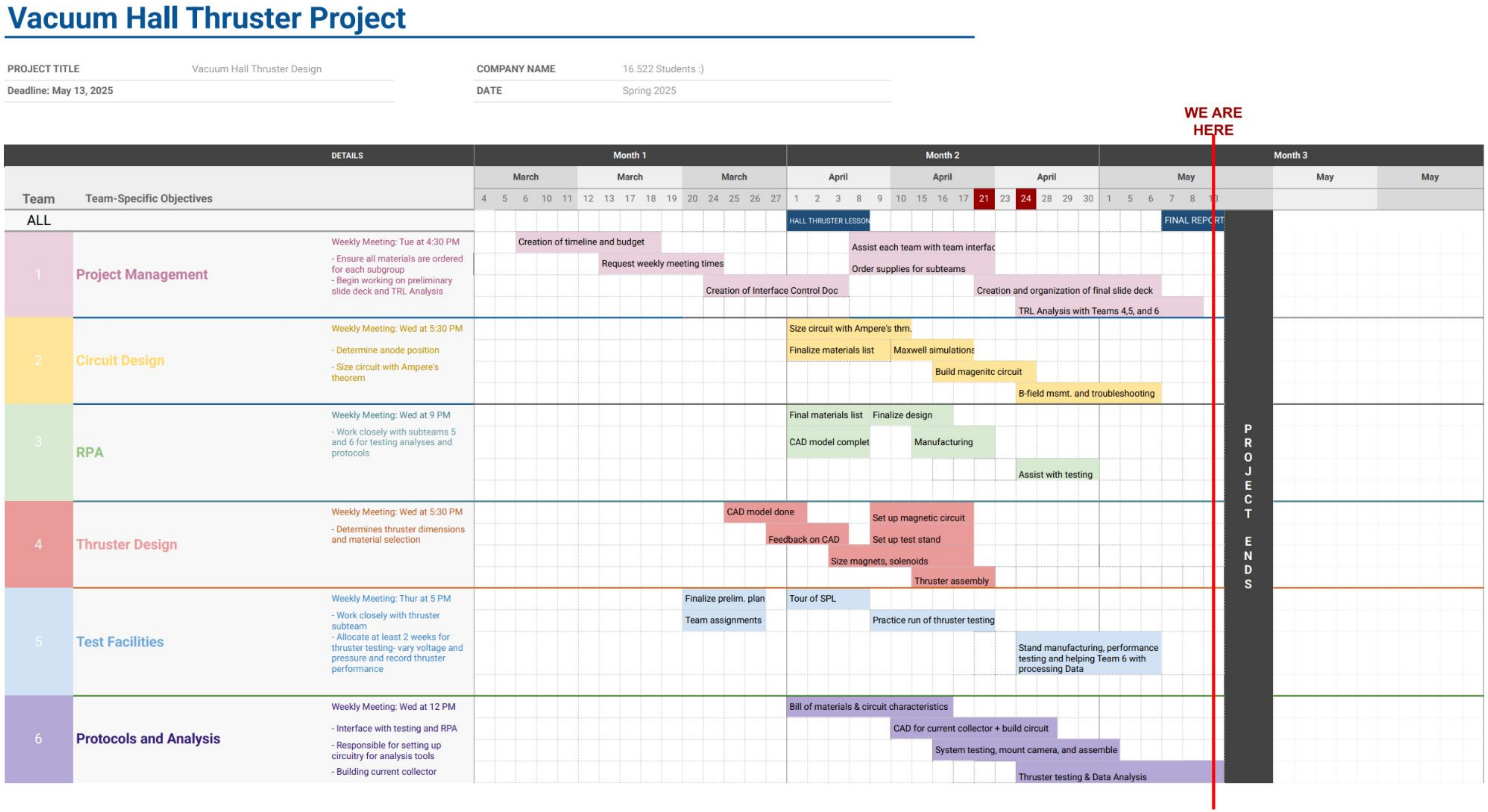
Used Google Drive and Slack to compile documentation and communicate across sub-teams





Project Timeline

Major Milestones and Deadlines



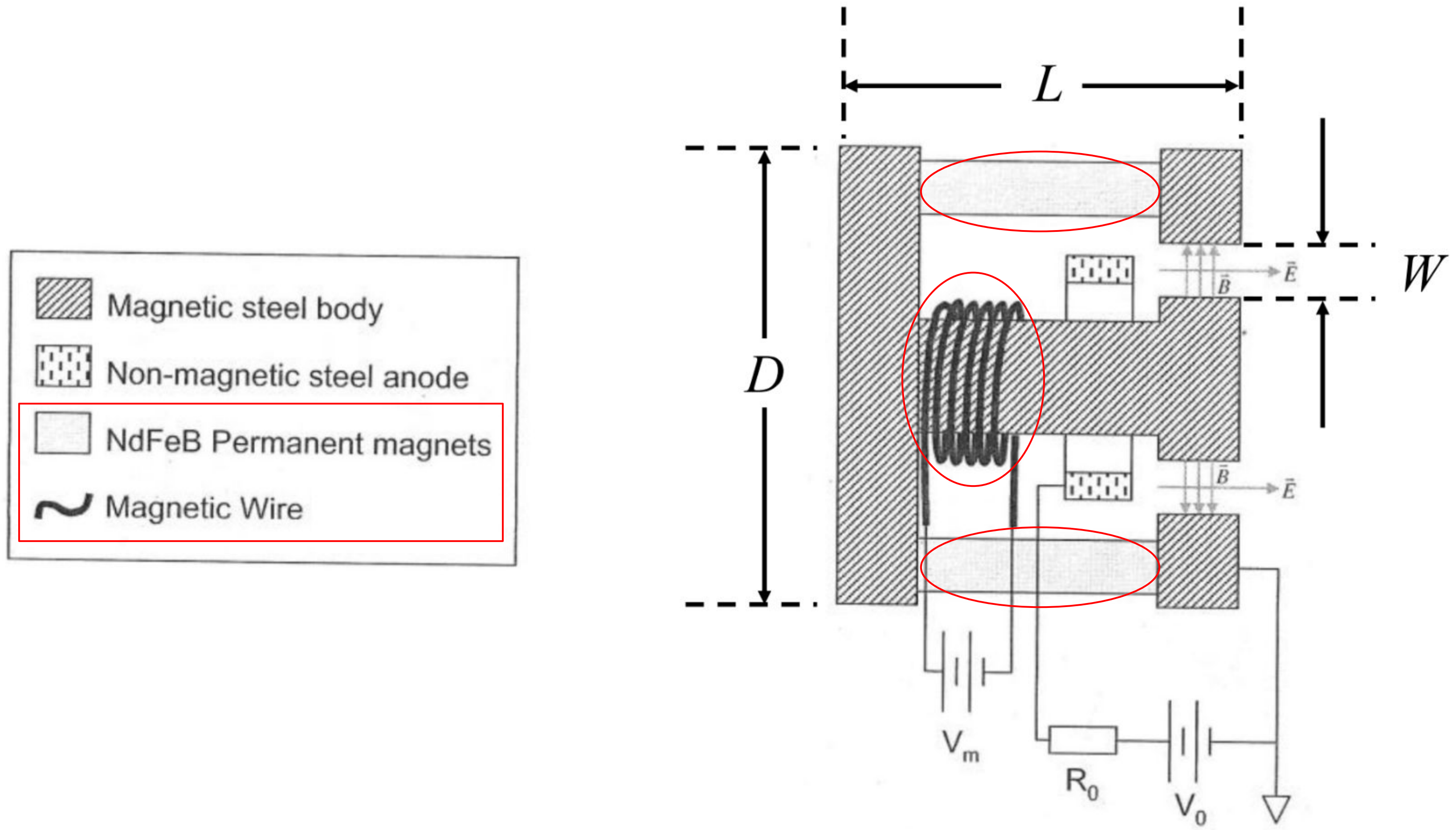
Project Budget

Subsystem	Purpose	Item	Description	Price	Quantity	Total	Link	Status	Total	$1,146.35
Thruster	Permanent Magnets	Neodymium Magnet	2" Long, 1/2" OD	$13.99	4	$55.96	https://www.apexm	Arrived		
Thruster	Anode	316 stainless steel stock	Sheet, 1/8" x 6" x 6"	$154.52	1	$154.52	https://www.mcma	Arrived		
Thruster	Front & Back Plates (Magnetic)	Wear-Resistant 410 Stainless Steel	sheet, 0.5" x 12" x 12"	238.2	1	238.2	https://www.mcma	Arrived		
Thruster	Front & Back Plates (teflon)	teflon sheet stock	sheet, 1/4" x 12" x 12"	$134.50	1	$134.50	https://www.mcma	Arrived		
Thruster	solenoid center	410 Stainless Steel Rod	rod, 1" diameter, 6" length	$14.98	1	$14.98	https://www.mcma	Arrived		
Thruster	bolts (for anode)	peek bolts	1/4-20, 3" length, peek	$9.69	6	$58.14	https://www.mcma	Arrived		
Thruster	bolts (for plastic and magnetic plate	316 stainless steel bolts	1/4-20, 1" length, 18-8 sta	$4.33	1	$4.33	https://www.mcma	Arrived		
Thruster	threaded rods (for sandwiching)	18-8 stainless steel threaded rods	1/4-20, 3.5" length	$13.75	1	$13.75	https://www.mcma	Arrived		
Thruster	lock nuts (18-8 stainless steel)	lock nuts	1/4-20, 18-8 Stainless Ste	$6.00	1	$6.00	https://www.mcma	Arrived		
Thruster	peek nuts	peak nuts	1/4-20, High-Strength Hig	$4.93	6	$29.58	https://www.mcma	Arrived		
Thruster	ptfe spacers (for anode)	PTFE Unthreaded Spacer	1/4" ID, 1/2" OD, 2" long	$12.78	4	$51.12	https://www.mcma	Arrived		
									Thruster	761.08
Retard…	RPA back plate	Copper plate		$48.09	1	$48.09	https://www.mcma	Arrived		
Retard…	RPA spacer	Alumina ceramic tube		$11.62	1	$11.62	https://www.mcma	Arrived		
Retard…	RPA grid frame	Aluminum sheet	0.016"x6"x12"	$5.91	1	$5.91	https://www.mcma	Arrived		
Retard…	RPA Mesh	Stainless Steel Wire Cloth		$15.42	1	$15.42	https://www.mcma	Arrived		
									Retard…	81.04
Testing	Mounting	L-brackets		$2.33	25	$58.25	https://www.mcma	Arrived		
Testing	Current collector sheets	Aluminum sheet	1/8"x12"x36"	$73.25	1	$73.25	https://www.mcma	Arrived		
Testing	Current collector spacers	PTFE spacers 1/4"		$9.29	10	$92.90	https://www.mcma	Arrived		
Testing	Data collection Microcontroller	Teensy 4.1		$31.08	1	$31.08	https://www.digikey	Arrived		
Testing	Thermocouple	K-type thermocouple		$15.24	1	$15.24	https://www.digikey	Arrived		
Testing	Thermocouple amplifier	Thermocouple amplifier		$14.95	1	$14.95	https://www.adafru	Arrived		
Testing	ptfe spacers	PTFE Unthreaded Spacer		$9.28	2	$18.56	https://www.mcma	Arrived		
									Testing	304.23

Magnetic Circuit Design

Sub-Team 2

Objectives and Scope



- The primary function of the magnetic circuit is **to generate a sufficient magnetic field to confine electrons** at the thruster exit.
- The Magnetic Circuit Team is responsible for:
  - **Designing magnet configurations** using both permanent magnets and magnetic wire
  - **Verifying anode position** with magnetic field profile
  - **Integrating these components** into the thruster assembly
  - **Conducting unit tests to verify that the magnetic field** strength is sufficient

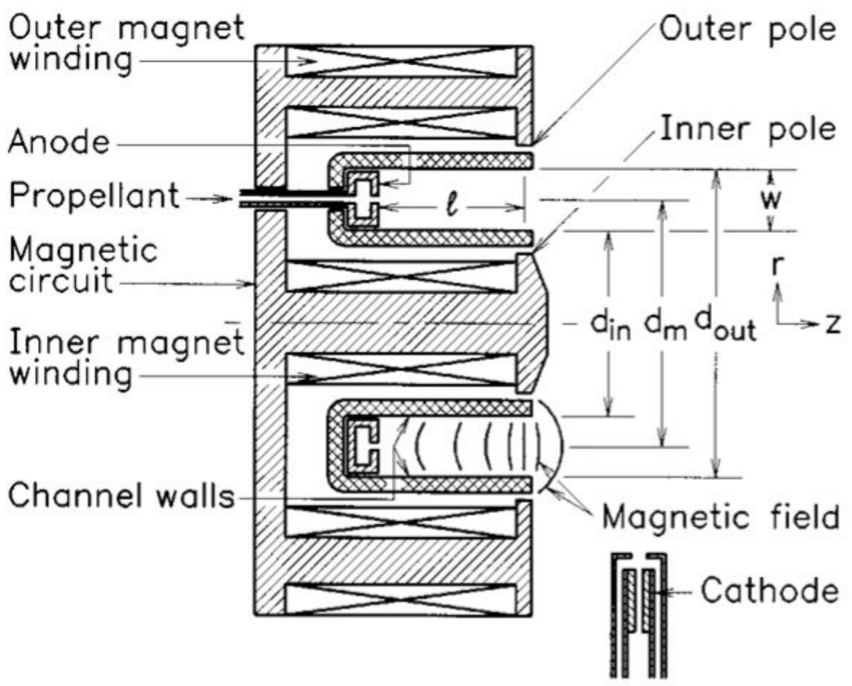


Figure 1. Closed drift thruster of the magnetic layer type.

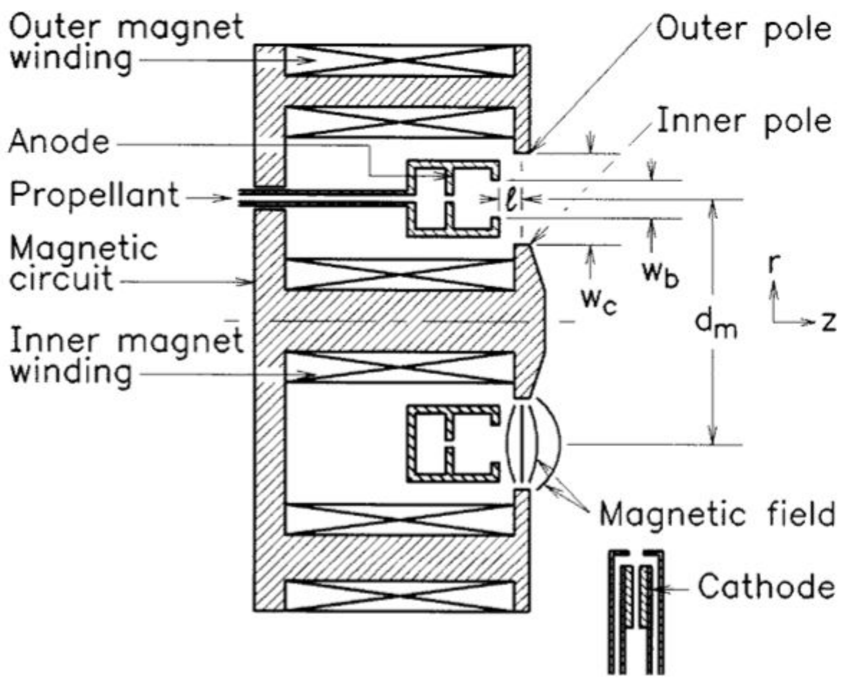


Figure 2. Closed drift thruster of the one stage, anode layer type.

Anode layer vs. Magnetic Layer

1) Verifying Anode Position

- Anode Layer vs. Magnetic Layer
  - Key differences in designs are the use of ceramic walls and the position of the anode (diagrams from Zhurin et. al[1])
    - Magnetic layer features ceramic walls and longer channel length
    - Collisions with ceramic walls cause lower electron temperature and thus slow acceleration process
    - Channel length affected by shape of magnetic field and lower electron temperature
  - Main consideration in deciding on anode layer was compatibility of ceramic walls with design requirements and simplicity

Design Results

1) Set Requirements

The requirements are set to ensure the Larmor Radius for electron is smaller than the channel length:

- Use the formulation given by Goebel and Katz[2]
- The electron temperature T_{ev} is estimated with the voltage at the anode[3] $V_b/10$.
- The minimum field $B_{\min}$ (-10% of the requirement) is given to satisfy the following conditions:
  - $w > 4r_e = 2$ (margin) $\times\ 2r_e$ (minimum required width) where $w = 1.2$ cm is the channel width

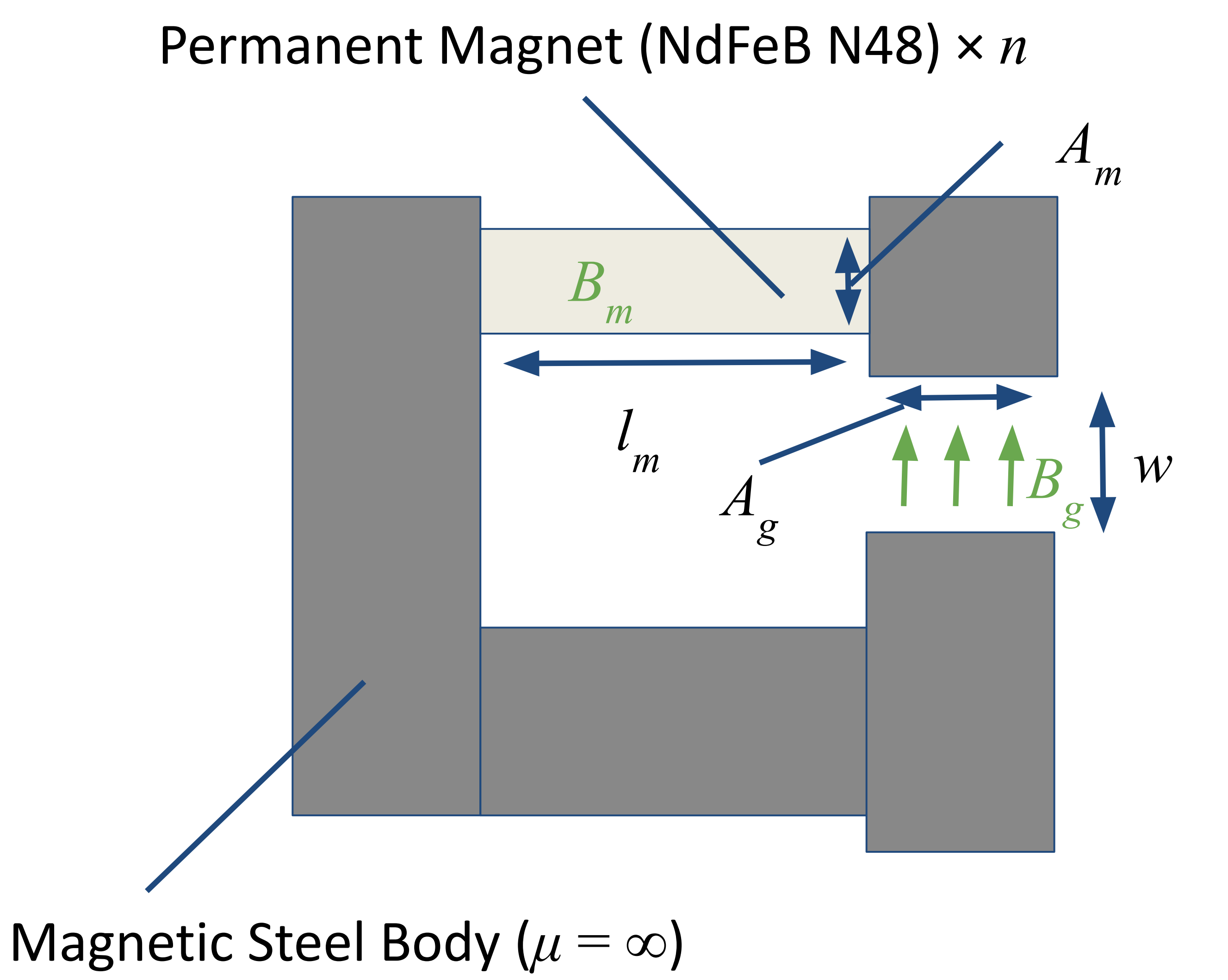
<u>B = 315 ± 31 (10%) [Gauss] is set as the requirement</u>

Note: Larmor Radius for ion with $B_{\min}$@5kV is $r_i = 227$ cm $>> w$

$$r_e = \frac{v_{th}}{\omega_c} = \frac{m}{eB}\sqrt{\frac{8kT_e}{\pi m}} = \frac{1}{B}\sqrt{\frac{8}{\pi}\frac{m}{e}T_{\mathrm{eV}}}$$

V [kV] @ anode	B requirement [Gauss]
1	141
3	244
5	315

$$r_i = \frac{v_i}{\omega_c} = \frac{M}{eB}\sqrt{\frac{2eV_b}{M}} = \frac{1}{B}\sqrt{\frac{2M}{e}V_b}$$



Ampère's Law: $H_m \boldsymbol{l_m} + H_g w = \boldsymbol{NI}$ (where $H_g = B_g/\mu_0$)

Continuity Equation: $B_m(\boldsymbol{n}A_m) = B_g A_g$

B-H Curve: $B_m = (B_r/H_c)\, H_m + B_r$ (where B_r and H_c are given)

Design Results

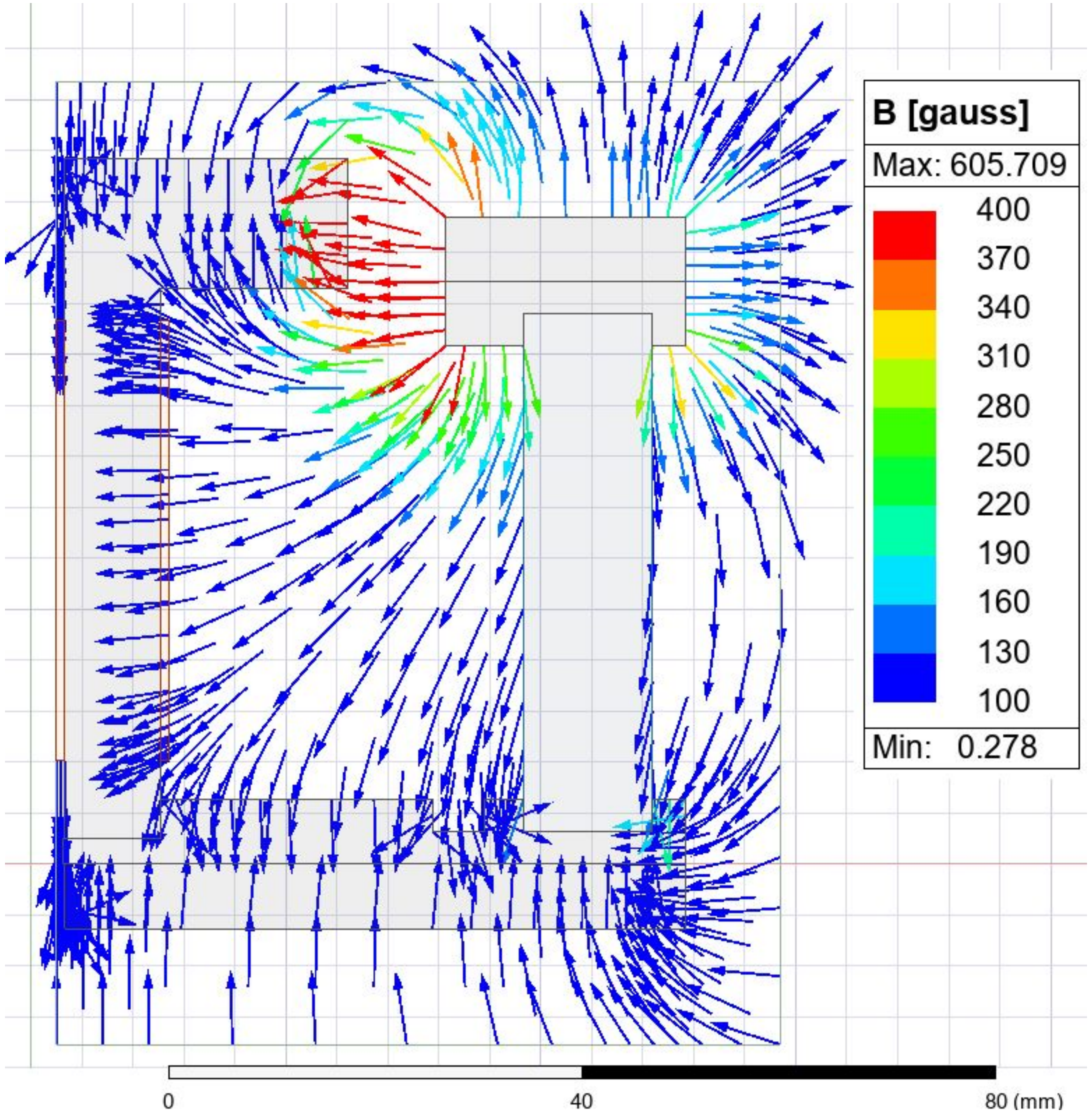
2) Magnetic Circuit Analytical Sizing

The number of permanent magnets (n), their length (l_m), the number of coil turns (N), and the applied current (I) tried to be determined using **Ampère's Law, continuity equations, and B-H curve**. The sizing process follows these steps:

1. Set the number (n) and the length (l_m) of permanent magnet (with $I = 0$)
2. Based on the magnet specifications, determine the required coil turns (N) and current (I)

Results and Limitation:

- The results (without current) indicate that achieving a magnetic field of 315 Gauss requires a magnet length of **only ~0.5 mm when using 4 NdFeB magnets** (½" diameter), which is physically unrealistic. To replicate the actual length (2"), the magnetic field needed to set **5.5~9.3 times stronger than the unit test value** of 263~445 Gauss. *2D simulation also overestimates by ~ 2 times
- The analysis assumes an infinite permeability for the steel body and **completely neglects magnetic field leakage**. Those would cause the overestimation of the calculated magnetic field strength.



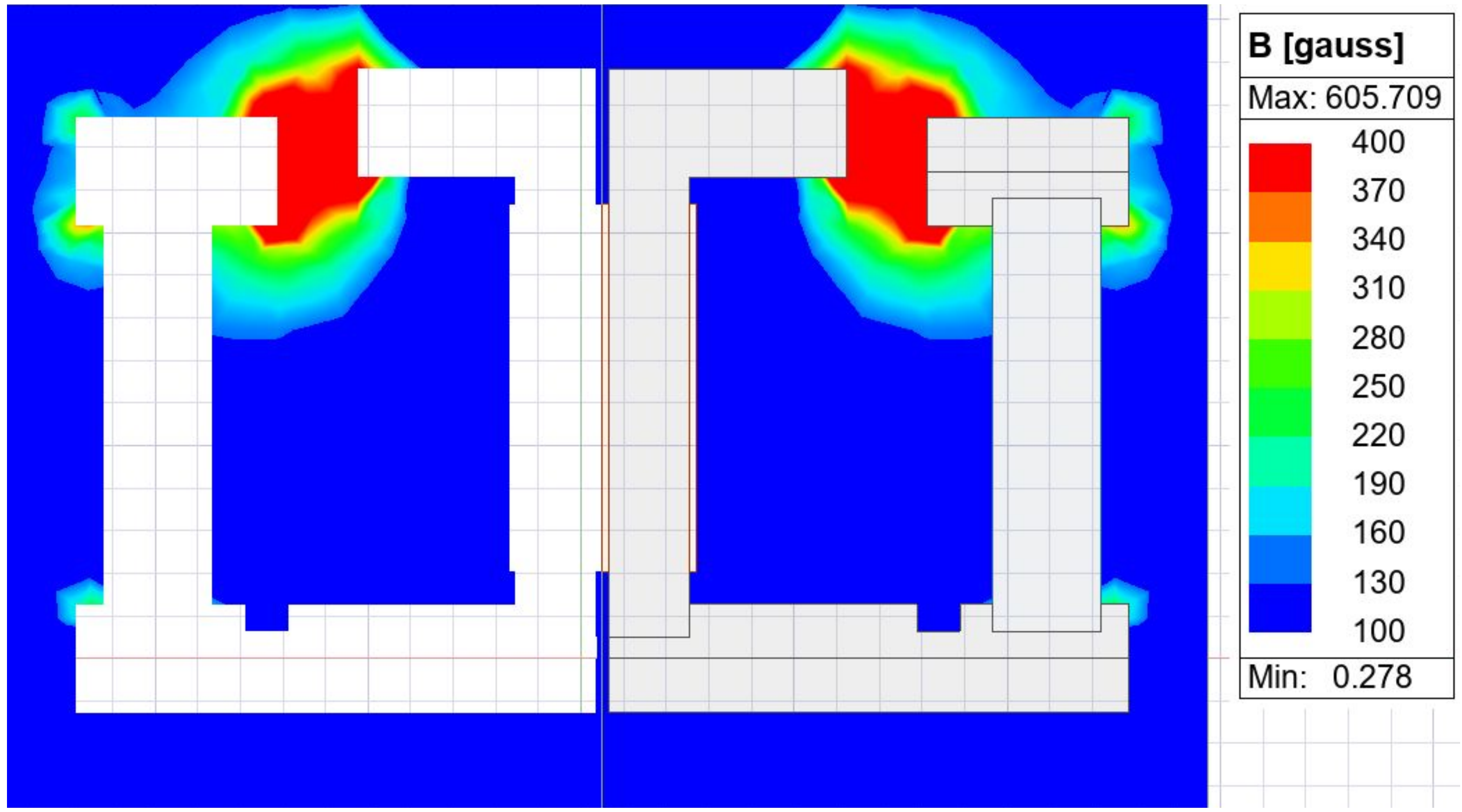
Design Results

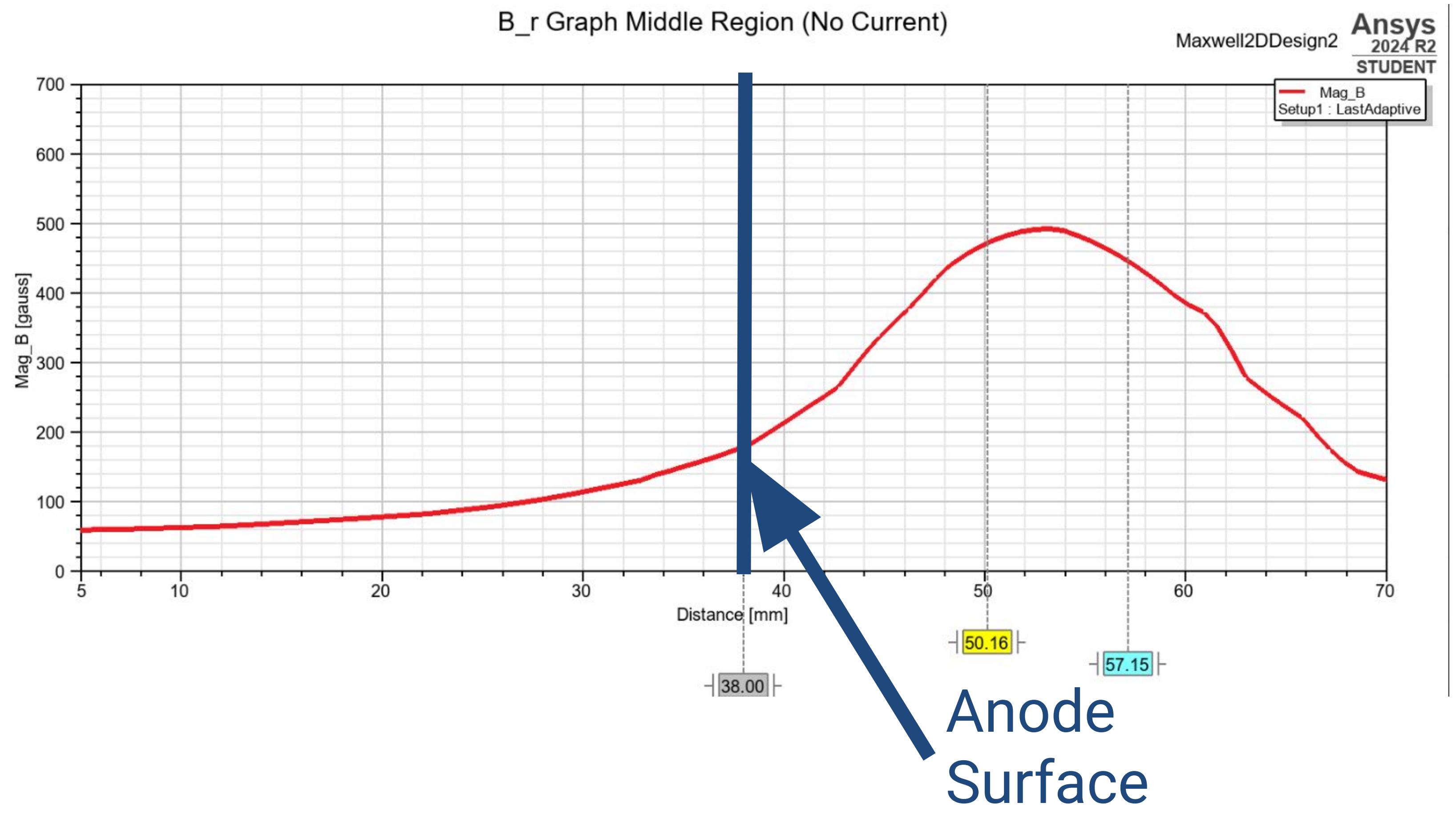
3) Maxwell simulation

The number of permanent magnets (n), their length (l_m), the number of coil turns (N), applied current (I), and anode placement have been validated using **2D Maxwell simulation**.

1. Simulation Setup
   a. B field sources
      i. **4** NdFeb magnets (2” × ½” diameter)
      ii. **100 turns** of 28 AWG copper wire w/ **0-3 A current**
   b. Odd boundary condition due to azimuthal symmetry
   c. 2D Projection of 3D Model
   d. Area Ratio due to simplified geometry[4]

2. Results
   a. Confirmed that necessary B field is met (w margin)
   b. Anode position





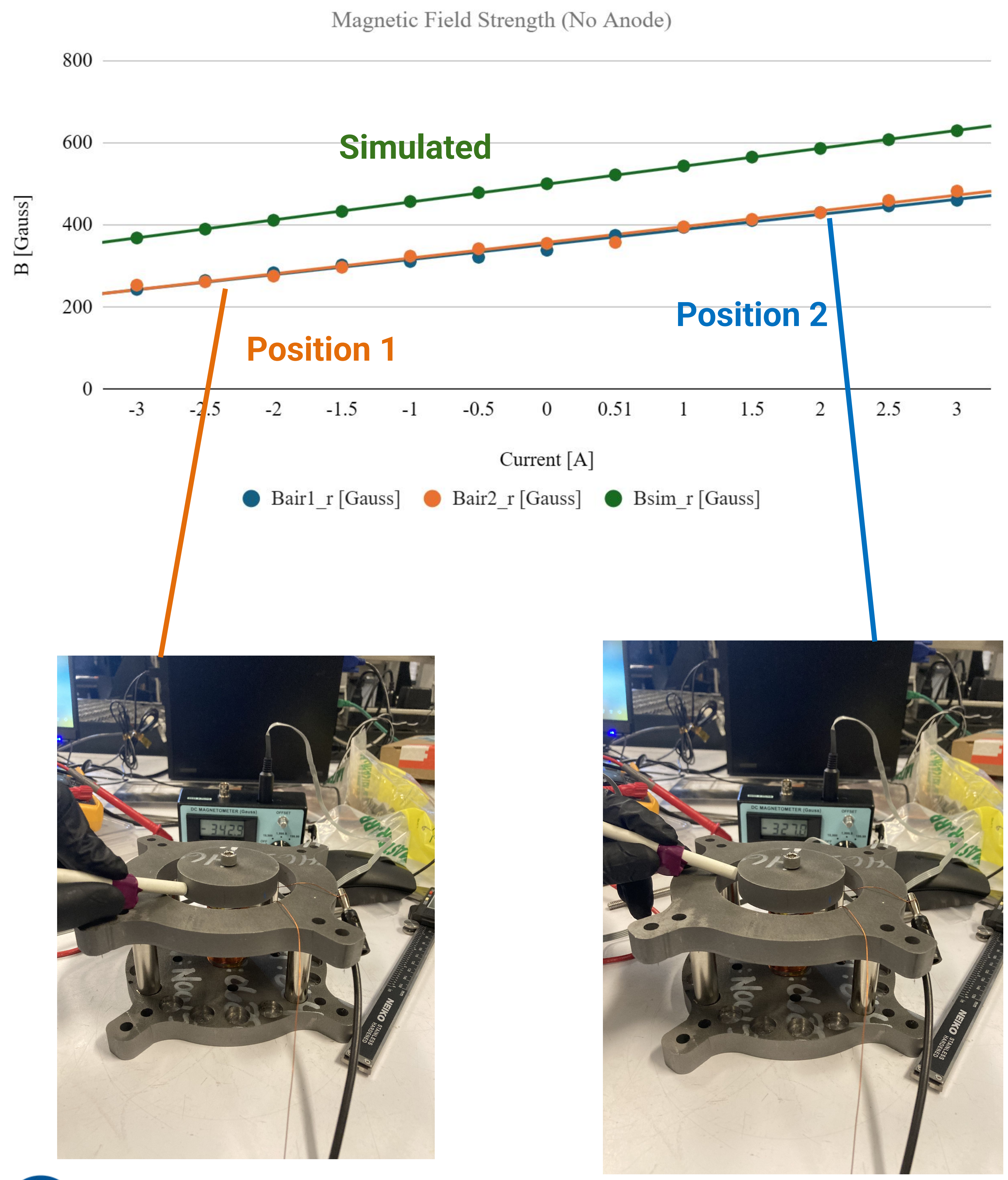
Design Results

3) Maxwell simulation

- Anode position
  - Factors affecting location decision:
    - Energy Efficiency/Performance:
      - Ideal magnetic field is delta function peaking at anode position[5]
      - The further away the bulk of the magnetic flux is from the anode, the less voltage ionized gases are accelerated through
      - Propellant ionizing behind the anode can be sent off in the wrong direction
    - Thermal:
      - Placing the anode to close to the peak in magnetic field causes electromagnetic heating that affects performance and damages the thruster

Construction

- Four permanent magnets satisfied the B-field requirements (plus margin) and ensured symmetry.
- **Two screw holes** were prepared to route the magnetic wire through the structure.
- Coil is **100 turns** of gauge 22 copper wire.
- Resistance: ~2.2 Ohm

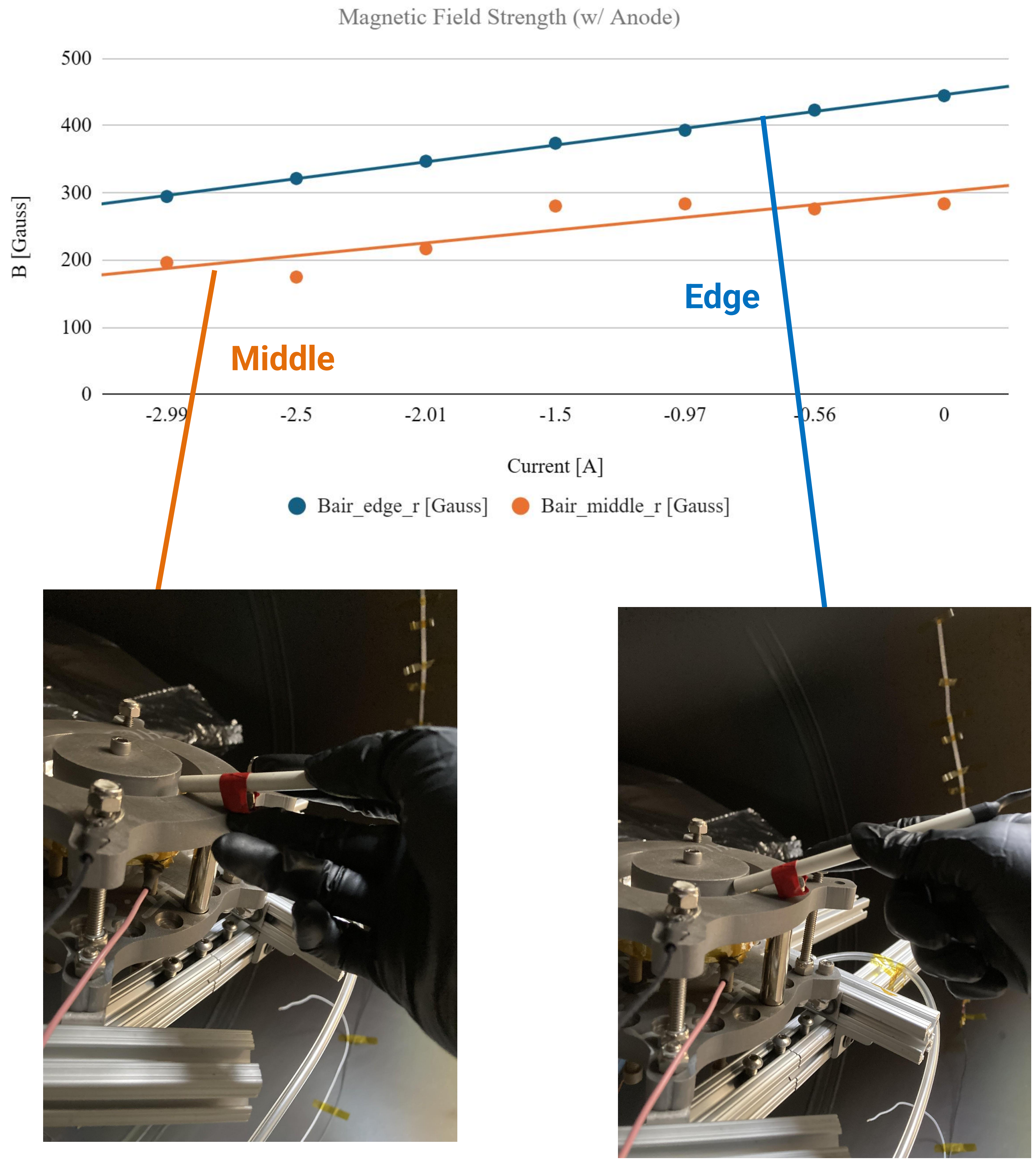


1st Unit Test Results

- The magnetic field was measured using a **Gauss meter** with the **full magnetic circuit configuration without the anode,** in two different azimuthal locations around the gap.
- Measurements were corrected for probe angle using: $B = B' / \cos(\theta)$, where θ = 15 deg.

- **Measured field range:** 250–460 Gauss =340-26%, 340+35%,
- **Value with no current:** 340 Gauss

- The results v.s. requirements **315 ± 31 Gauss (±10%)**
- Chose larger magnetic field



2nd Unit Test Results

- The magnetic field was measured using a **Gauss meter** with the **full magnetic circuit configuration and anode**
- Limited measurement time due to wiring issues, data only ranges from reverse 3 amps to 0 amps
- Measurements were corrected for probe angle using:

  $B = B' / \cos(\theta)$, θ = 21 deg (edge), 22 deg (middle).
- **Estimated (using slope of trendline) field range:**
  - Middle: 182 - 397 Gauss (-54%)
  - Edge: 295 - 595 Gauss (-50%)
- **Measured Value with no current:**
  - Middle: 263 Gauss
  - Edge: 445 Gauss
- Meets B field requirements **315 ± 31 Gauss (±10%)** in both regions

Test Results

Magnetic Circuit Current (A)	RPA Voltage (kV)
0	0.58
0.2	0.61
0.4	0.66

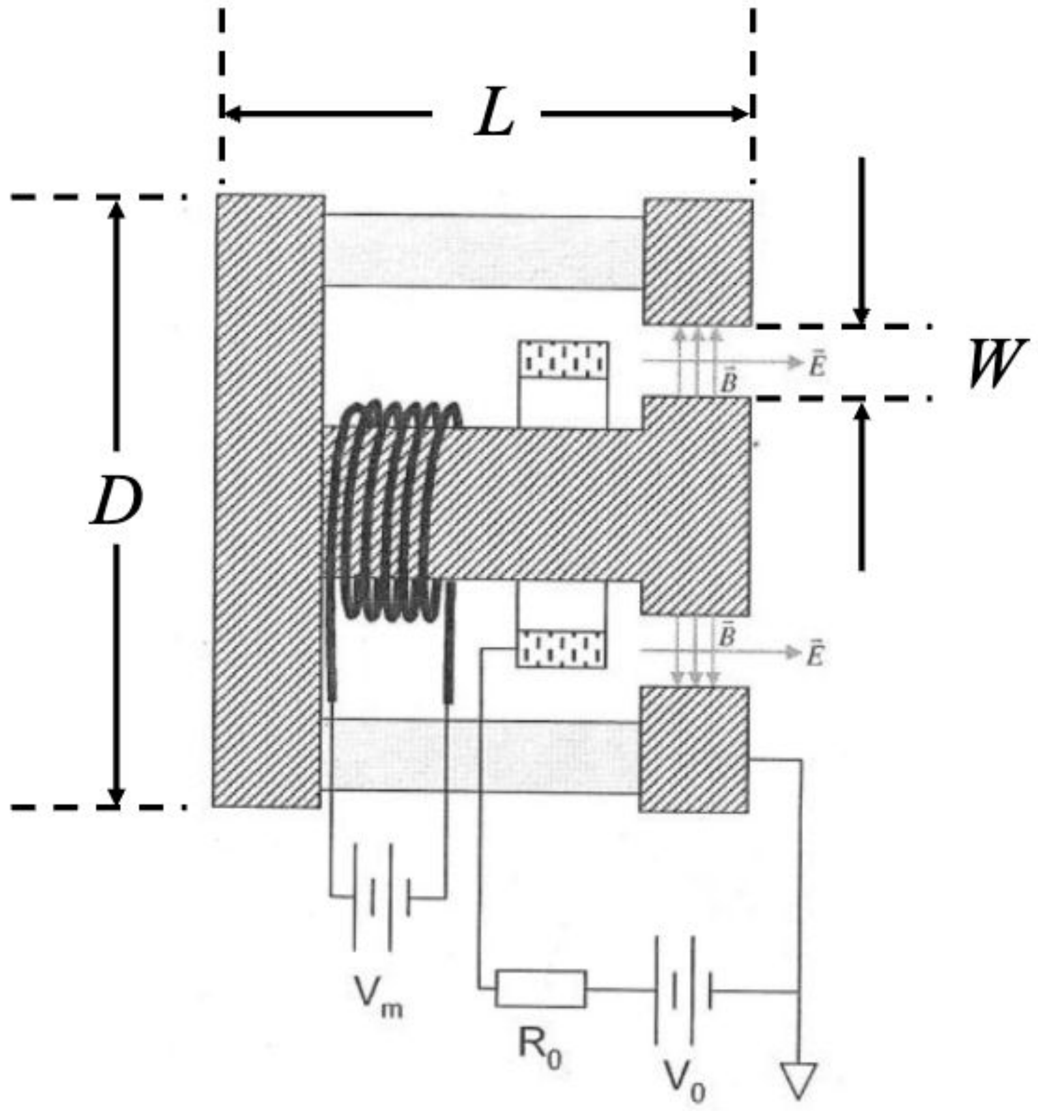
- Tested at -3 to 3 A
- Thruster turns off at less than -1 A
- Missing data due to RPA sizing

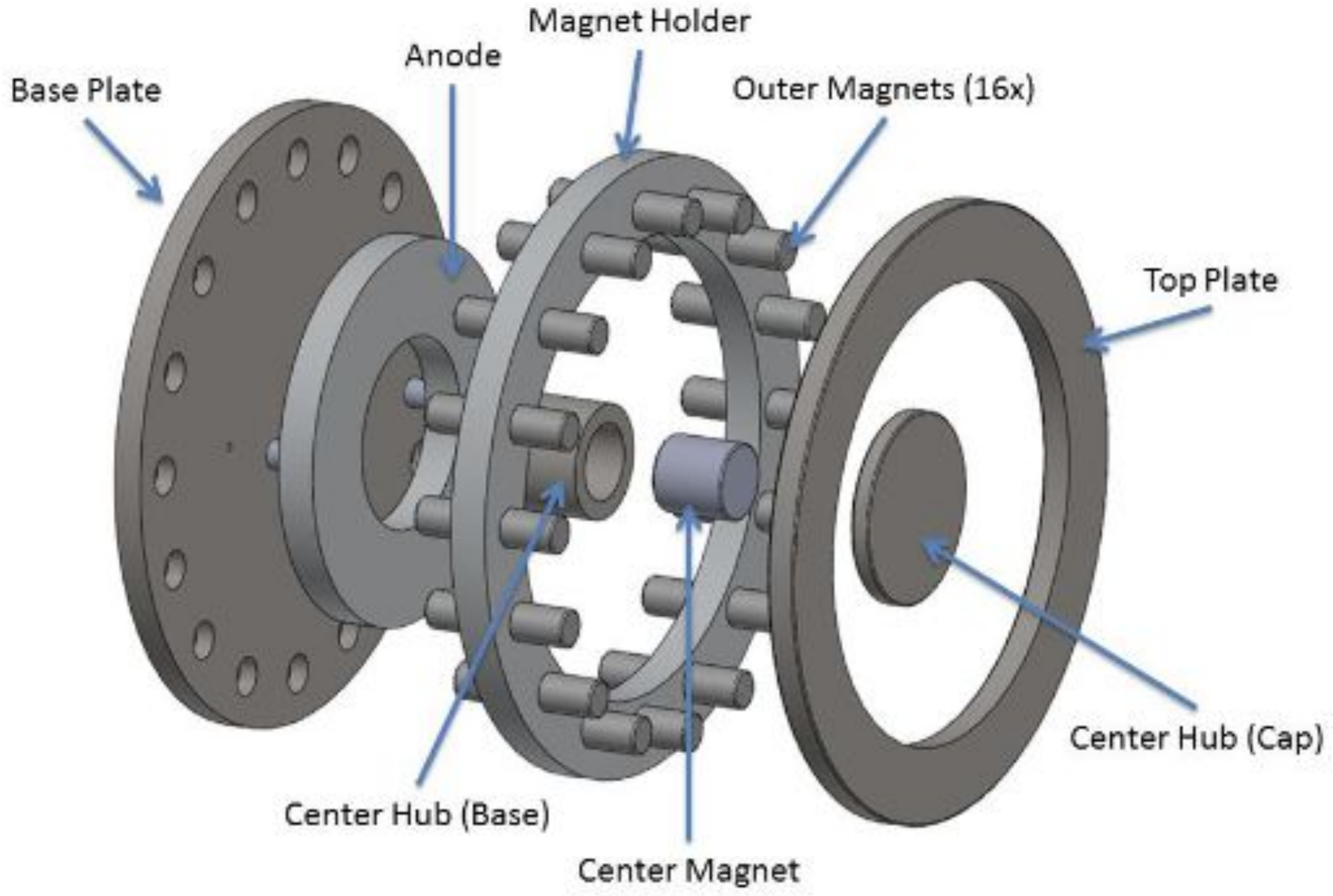
Thruster Design & Construction

Sub-Team 4

Objective & Scope

- Design and Build a Vacuum Hall Thruster in accordance with the project standards





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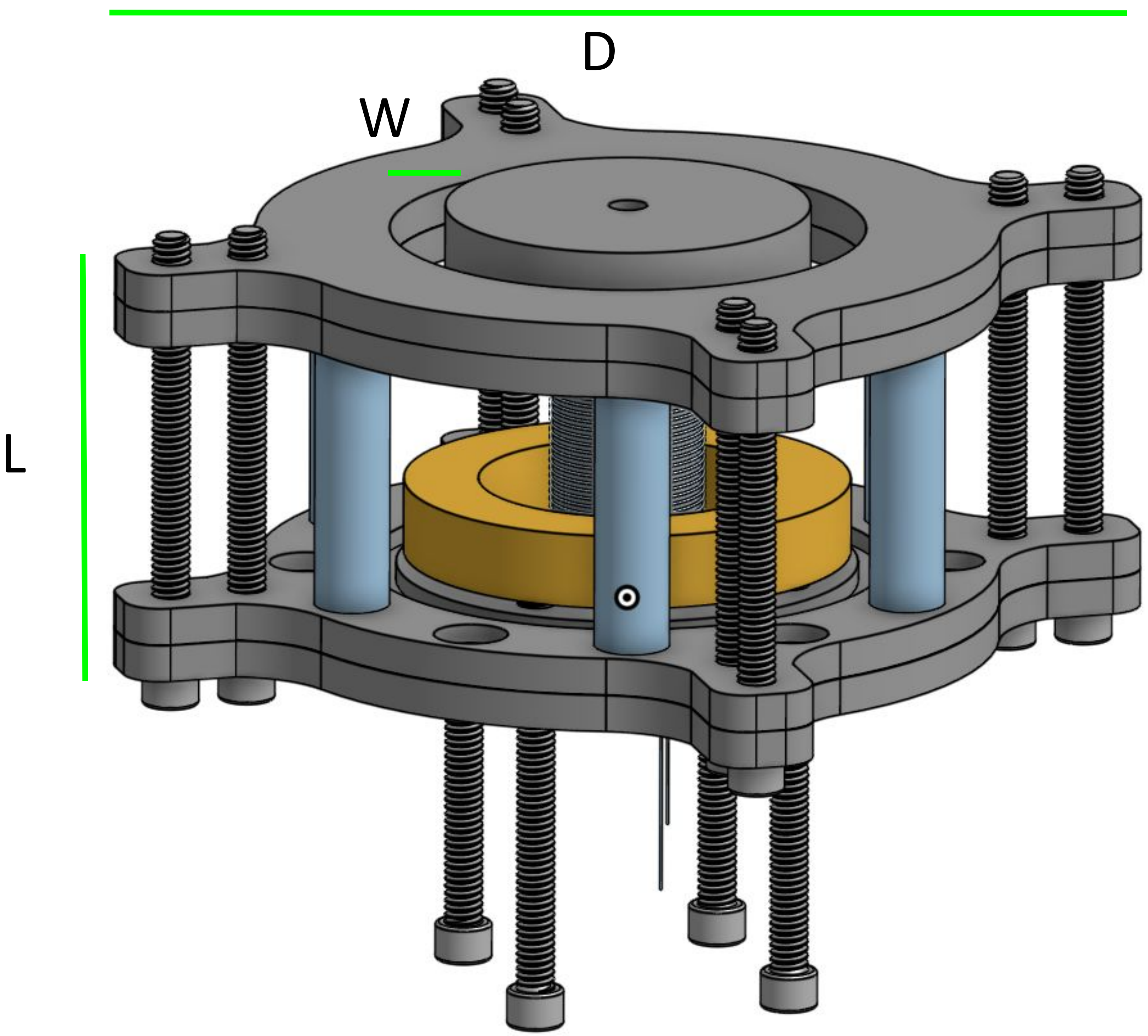


Design Phase

Key Constraints

Project Requirement	Our Design
L < 15 cm	L = 7 cm
8 < D < 15 cm	D = 13 cm
0.5 < W < 1.5 cm	W = 1.2 cm

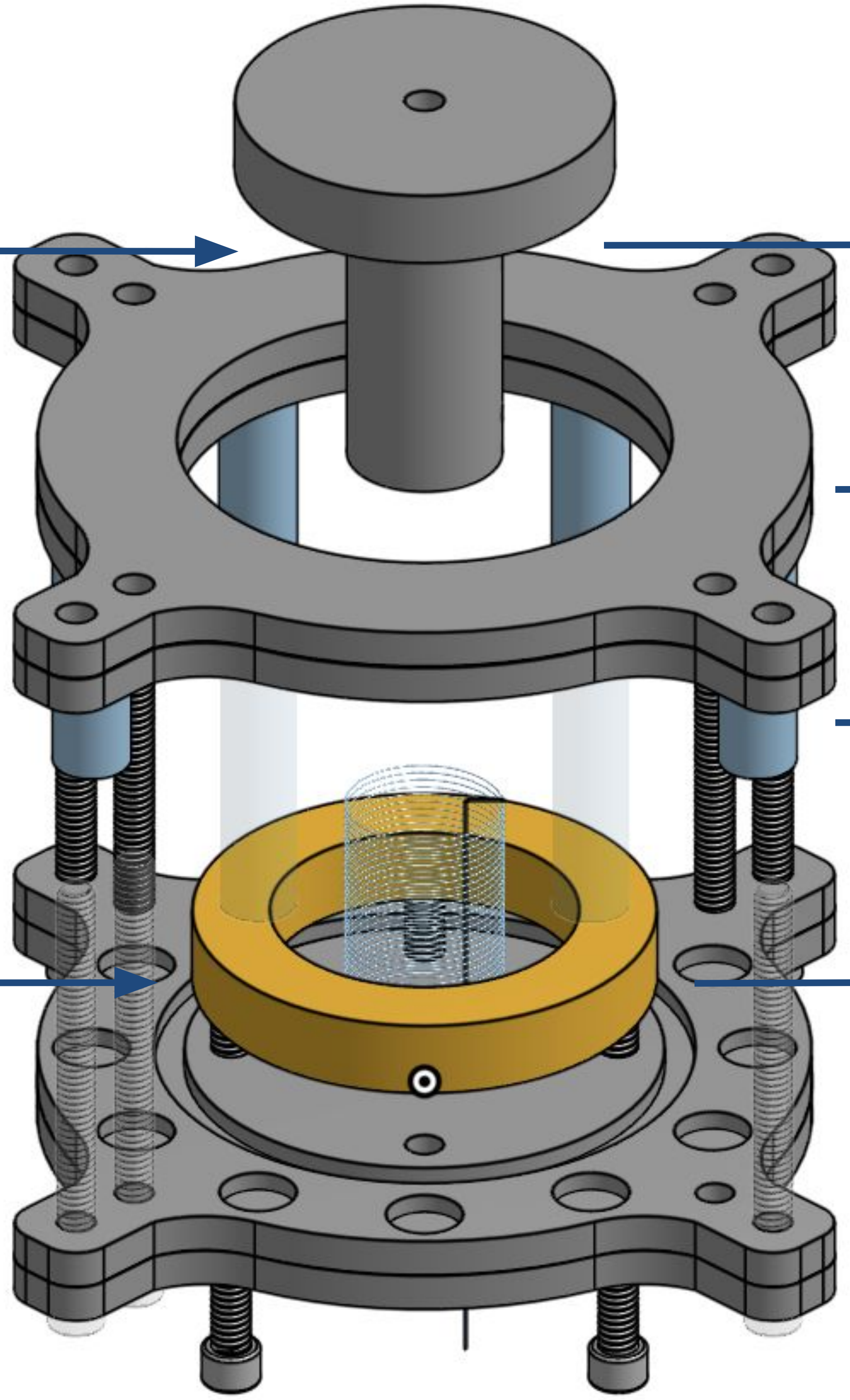
Notable Choices

1. Adjustable position of the anode
2. Extra pockets for additional permanent Magnets
3. Removable core top to adjust width

All Design was completed in Onshape

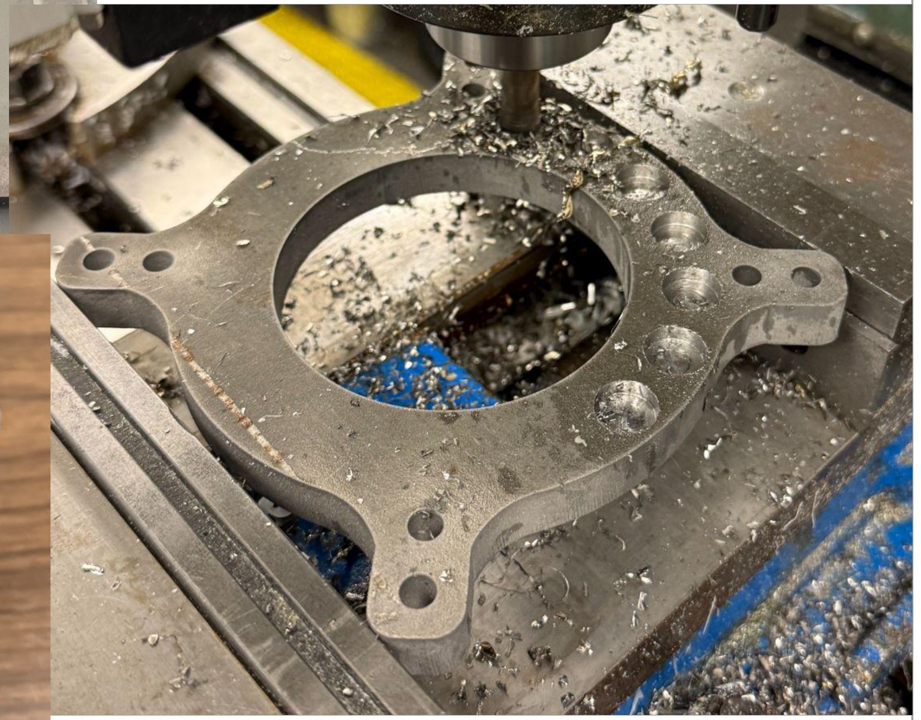
Component List

Material Selection

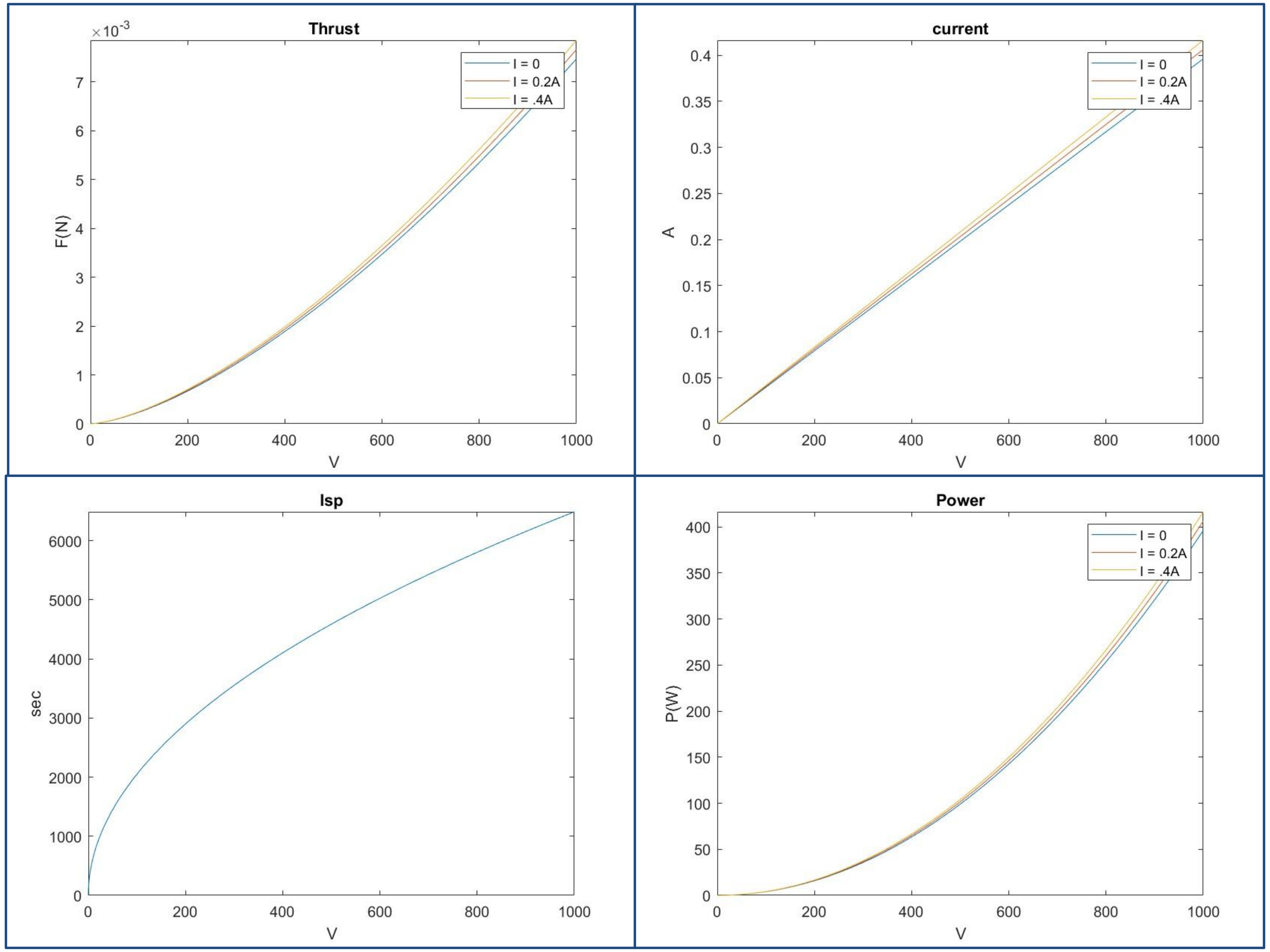


Component	Material
Core	410 Magnetic Stainless Steel
Top Plate	410 Magnetic Stainless Steel
Permanent Magnets	Neodymium
Anode	6061 Aluminum
Base Plate	410 Magnetic Stainless Steel

Manufacturing Overview

Component Order	Tools Used	Images
1. Top plate, Bottom plate, Core top	Water Jet	
2. Permanent Magnet Pockets	Mill	
3. Wiring holes in Core	Mill, Lathe	
4. Anode	Water Jet, Mill	

Performance Predictions



Equations:

- Thrust: $F \approx \dot{m}_i \bar{c}_i = 2\varepsilon_0 \alpha \sqrt{\frac{2e}{m_i}} n_n Q_i \eta_i^{1/2} \eta_E^{1/2} \gamma^{1/2} B_r V_a^{3/2} A$
- Current: $I_B^i = j_B^i A = \frac{2\varepsilon_0 e}{m_e} n_n Q_i \eta_i^{1/2} \gamma^{1/2} B_r V_a A$
- ISP: $I_{sp} \approx \frac{1}{g} \sqrt{\frac{2e}{m_i}} \eta_E^{1/2} V_a^{1/2}$
- Power: $P_a = I_a V_a = \frac{2\varepsilon_0 e}{m_e} n_n Q_i \eta_i^{1/2} \gamma^{1/2} B_r V_a^2 A$

Variables:

- n_n = neutral density 3.2*10e19
- η_i = ionization efficiency 0.771
- η_e = energy efficiency 0.6
- Q_i = ionization cross-section 9.915e-20
- γ = electron energy 0.5
- B_r= radial B field (38.426 * I + 302.32)/10000;
- A = Area of channel 0.0021

Key Takeaways:

- We should be able to observe a subtle increase in our output from the variable B field
- This is assuming we capture all the current with the current collector

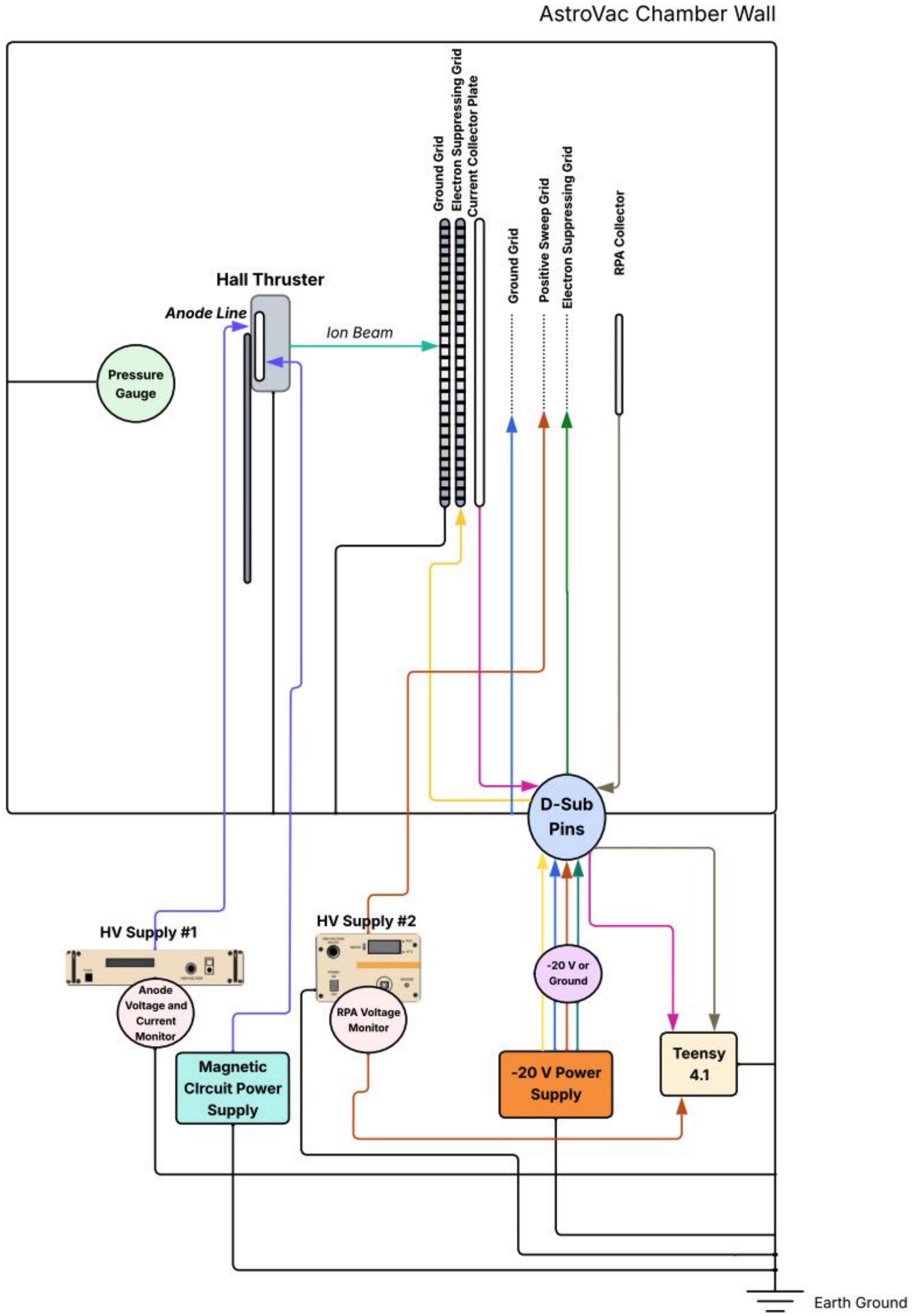
Testing Design and Protocols

Sub-Teams 5 & 6

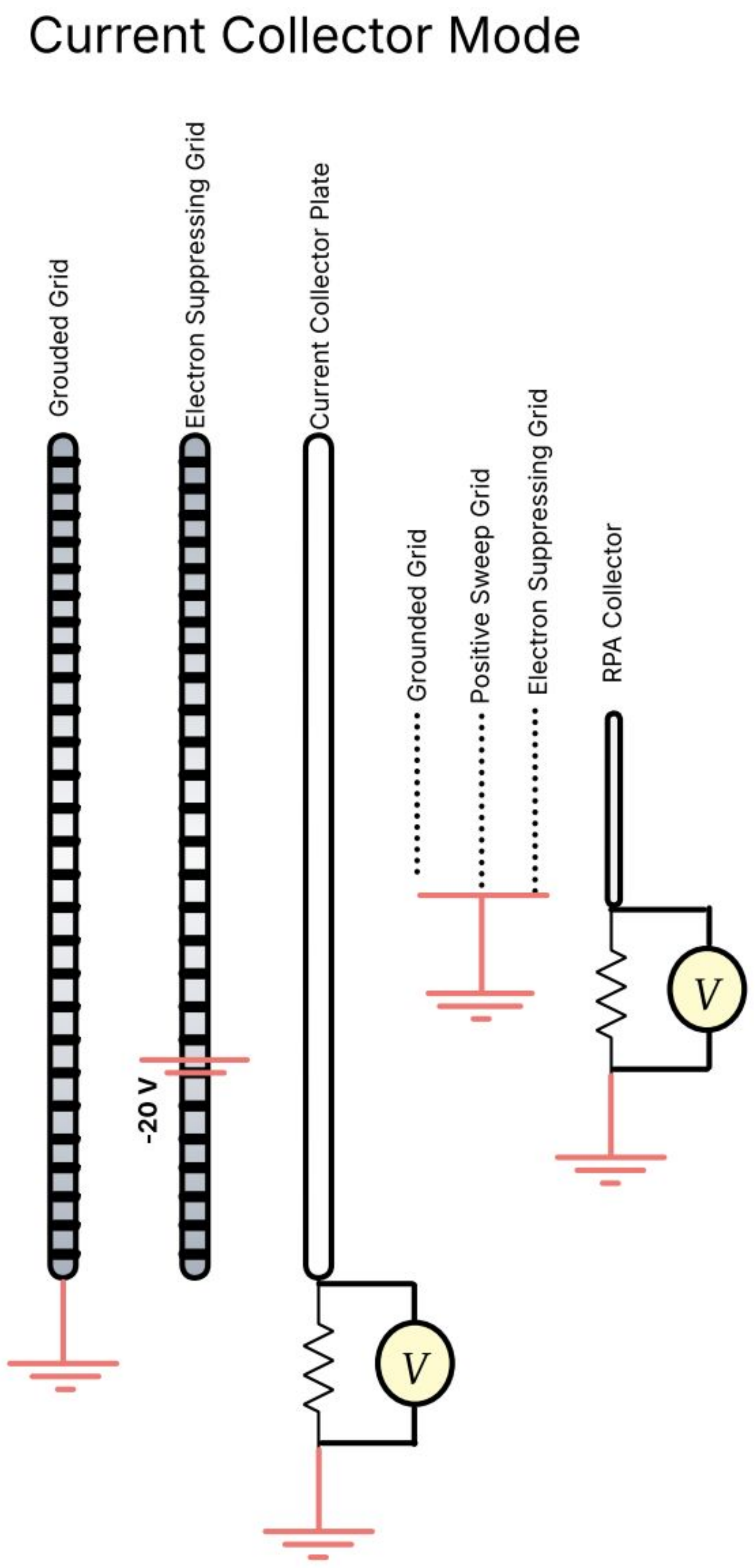
Sub Team Objectives

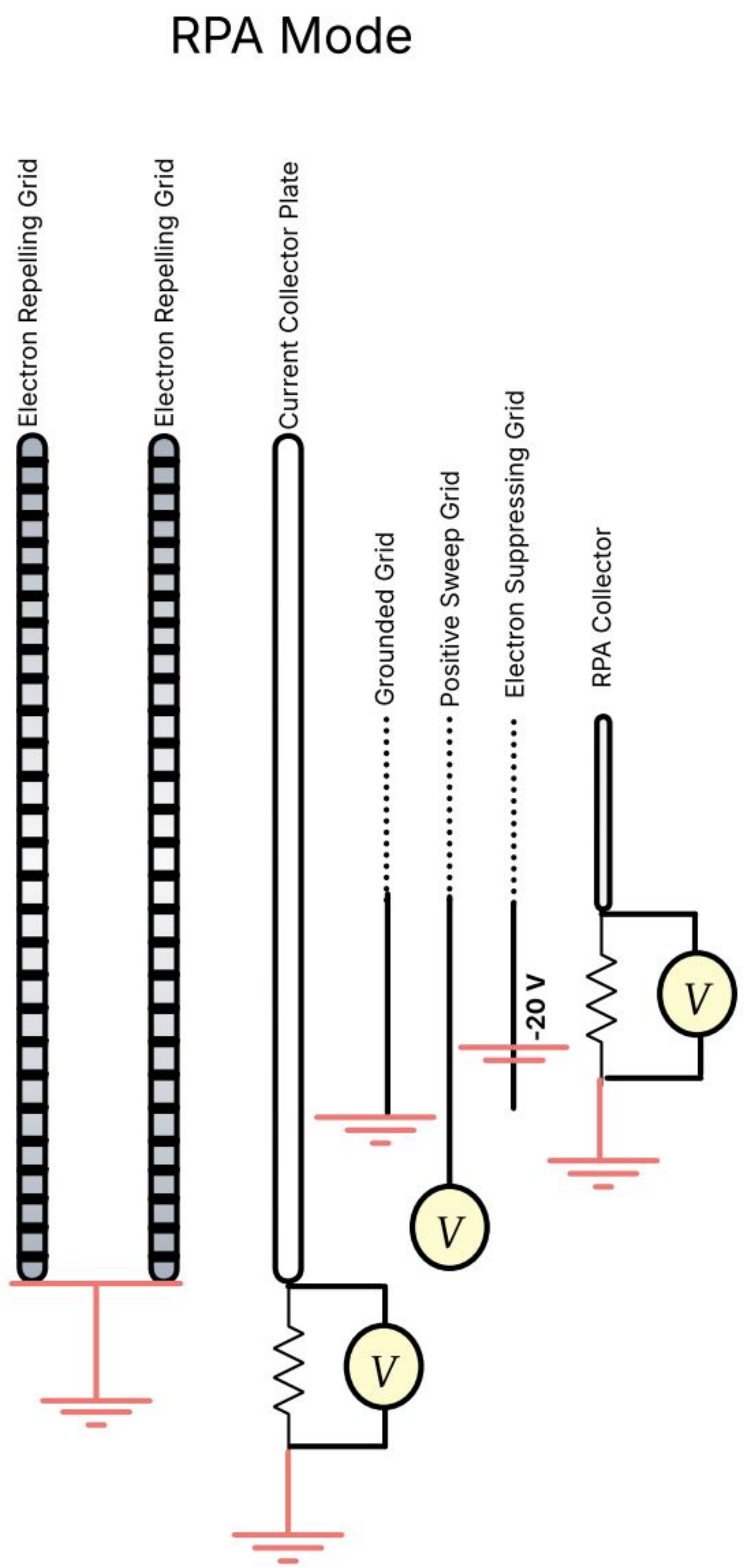
- Design and assemble full test infrastructure
- Develop all electrical wiring and sensor integration
- Design and built diagnostics (RPA, current collector)
- Build DAQ system and real-time GUI for data logging
- Write SOPs and ensure testing safety

Electrical Connections



**HV Power Supply # 1 = Spellman's Bertan brand of 205B series High Voltage Power Supply
**HV Power Supply # 2 = Spellman's Bertan brand of 230 series High Voltage Power Supply





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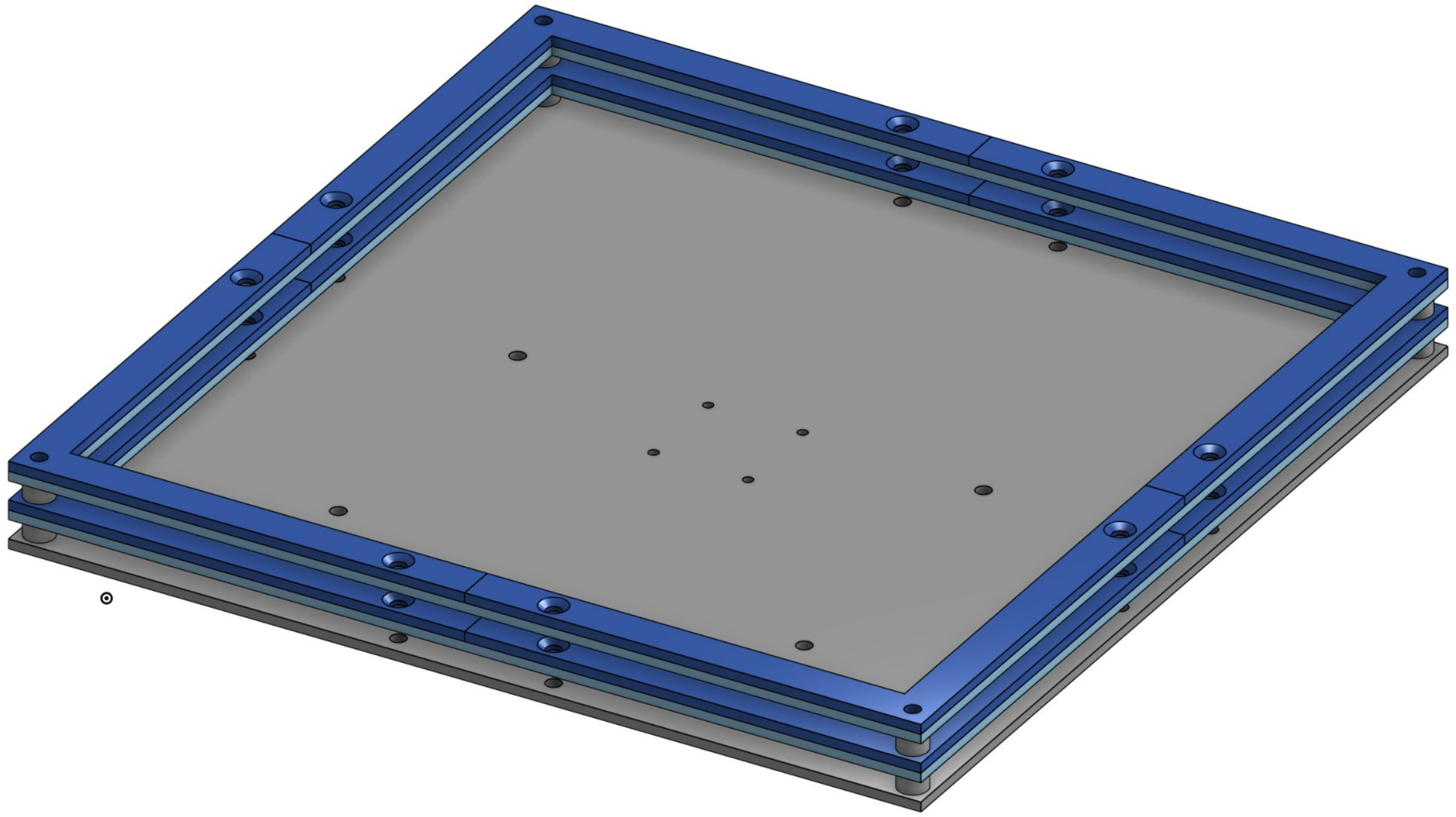
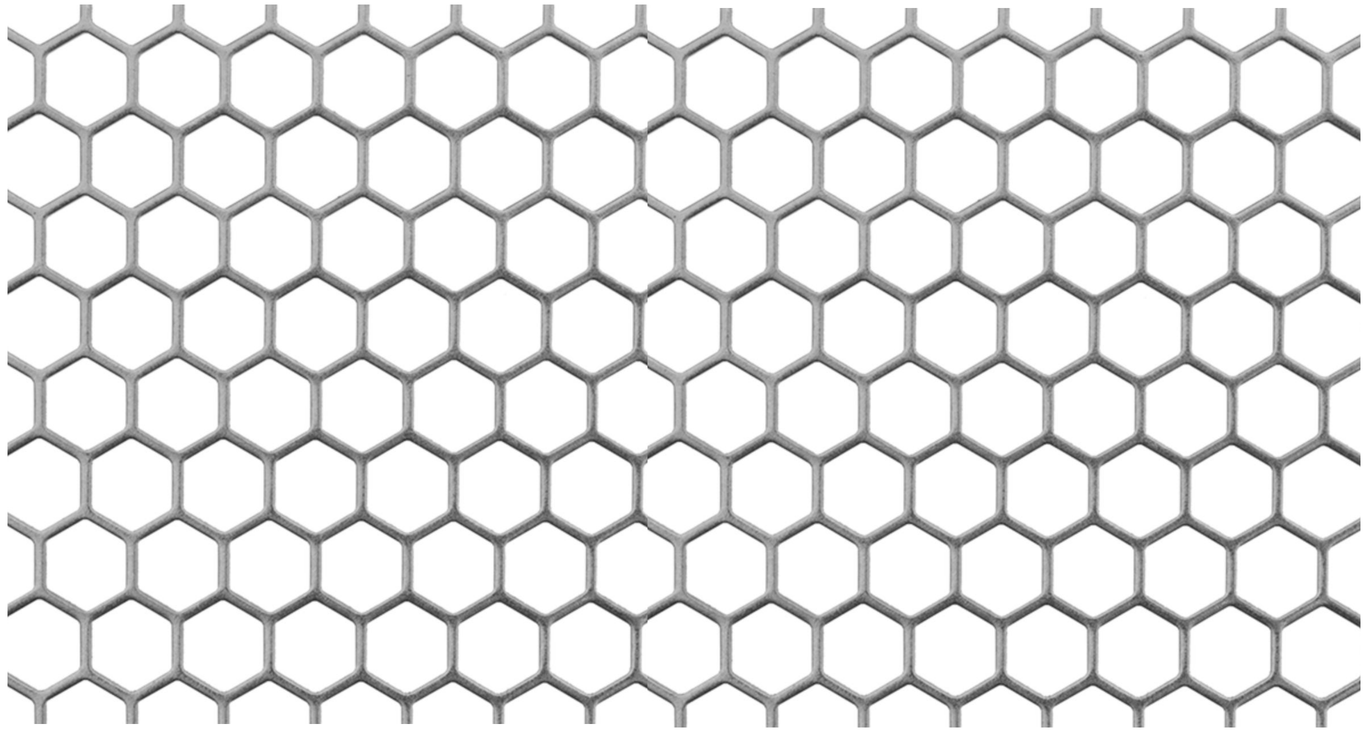
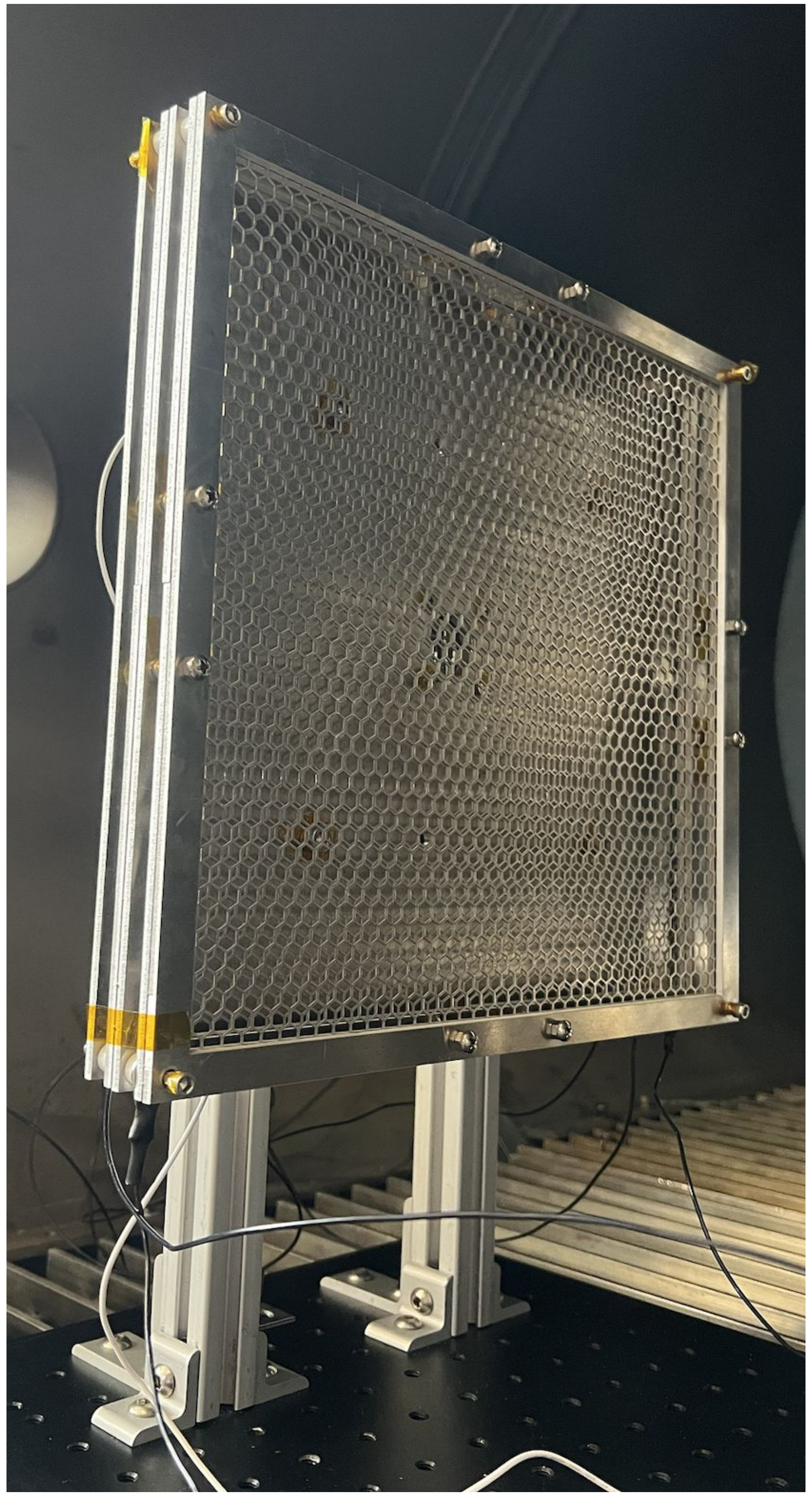
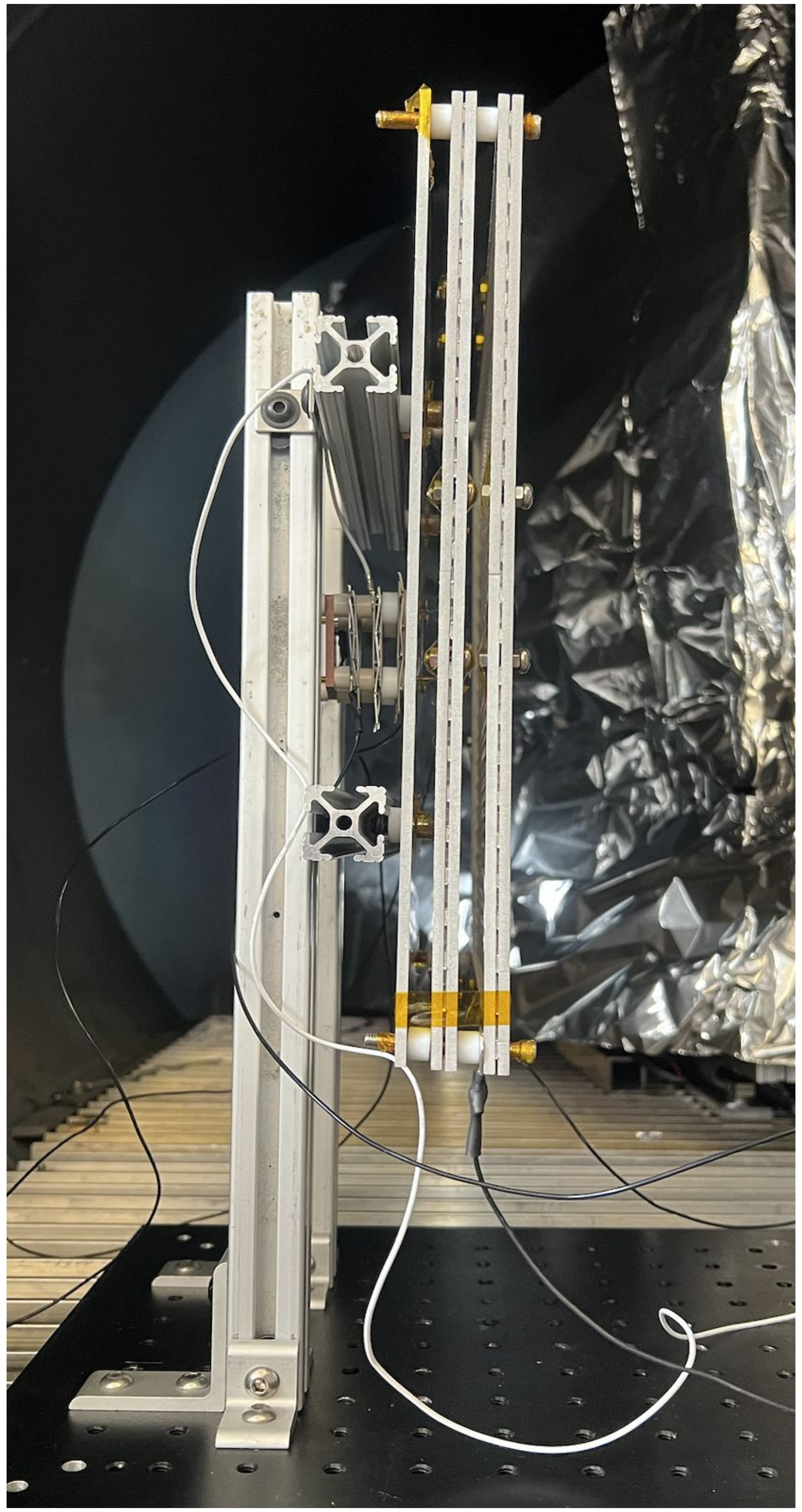
Current Collector



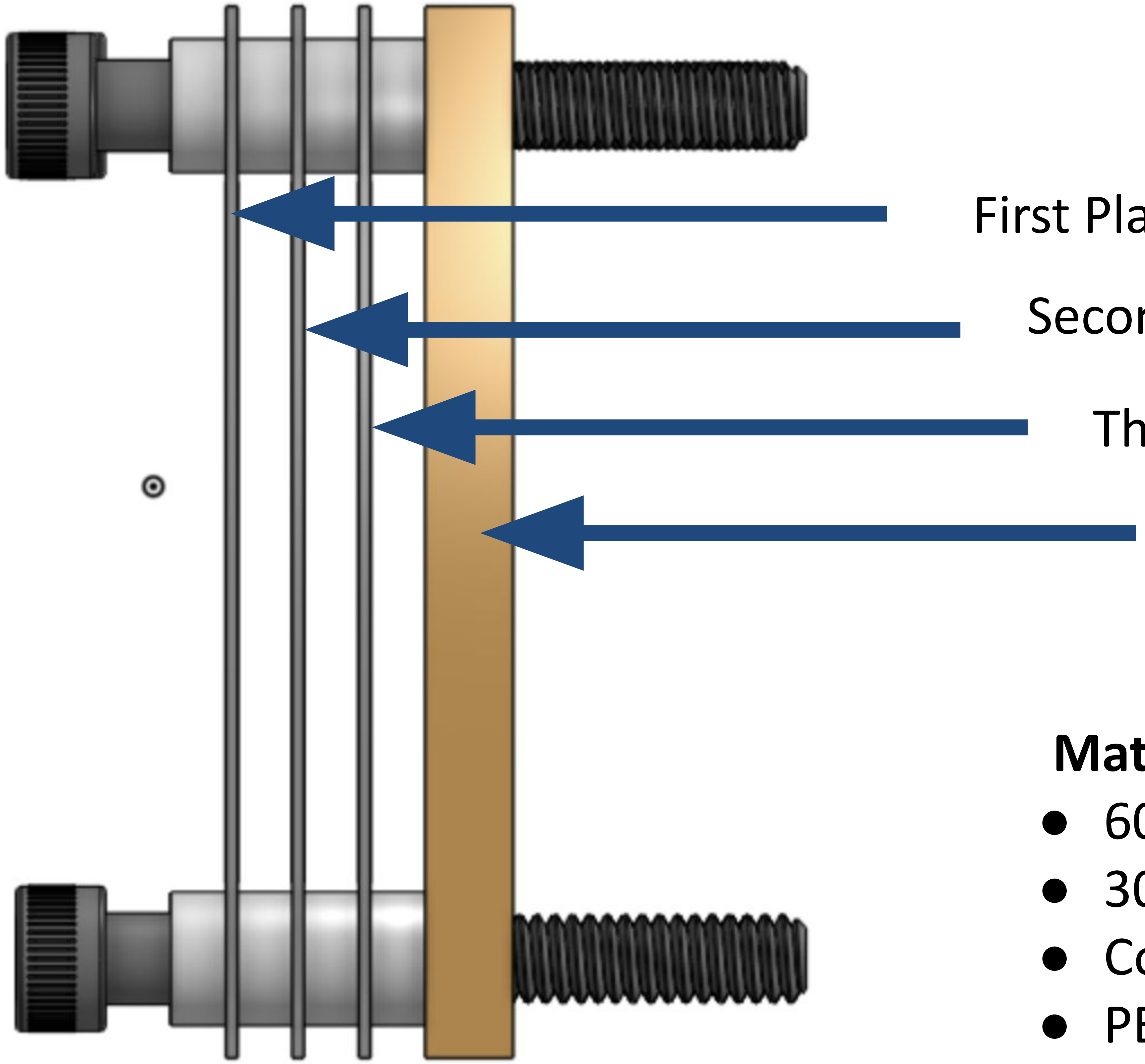






[1] Ferda, B. (2015, September 24). *Retarding potential analyzer theory and design*. Princeton University. https://w3.pppl.gov/ppst/docs/ferda.pdf

RPA Design



First Plate (Grounded)

Second Plate (Positive Sweep: 0 - 1.3Va)

Third Plate (Static Negative: -20 V)

Collector

Materials Used

- 6061 Aluminum
- 304 Stainless Steel Wire Cloth
- Copper Plate
- PEEK Spacers

RPA Design

Debye Length

$$\lambda_d = \sqrt{\frac{\epsilon_0 k_b / q_e^2}{n_e/T_e + \sum_{ij} z_{ij}^2/T_i}} \approx \sqrt{\frac{\epsilon_0 k_B T_e}{n_e q_e^2}}$$

Optimal Spacing

$$x = 1.02\lambda_d(\frac{eV_d}{k_B T_e})^{\frac{3}{4}}$$

Conservative Estimate for Grid Spacing

$$x \approx 4\lambda_d$$

With

$$n = 10^{14}\ \mathrm{electrons/m^3},\ T = 3\ \mathrm{eV}$$

we get

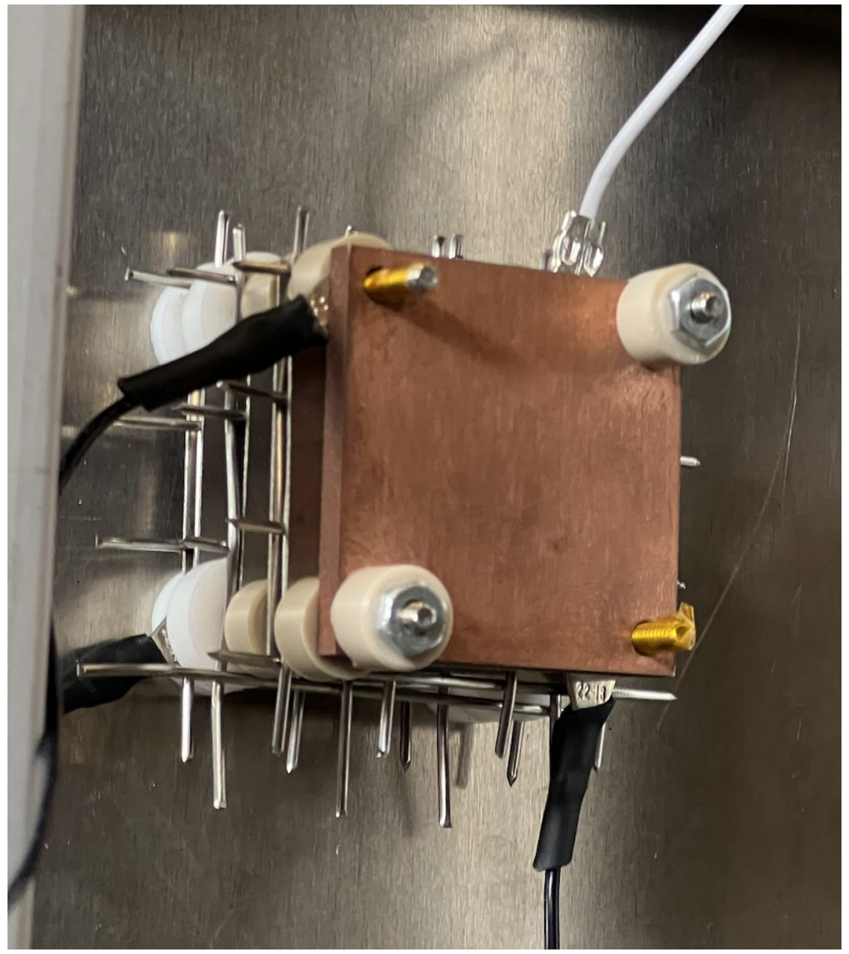
$$\lambda = 1.28\ \mathrm{mm}$$

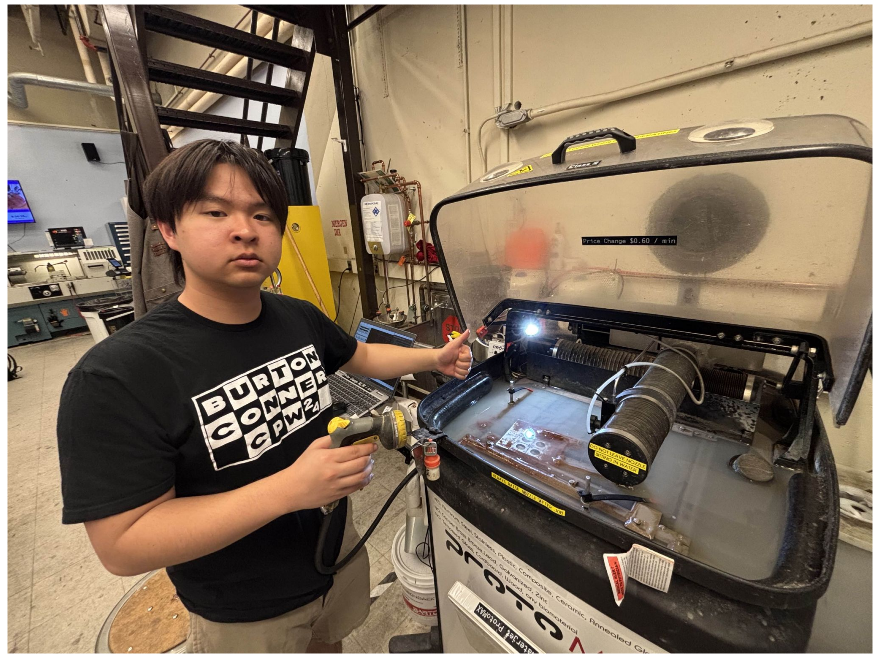
which makes our grid spacing

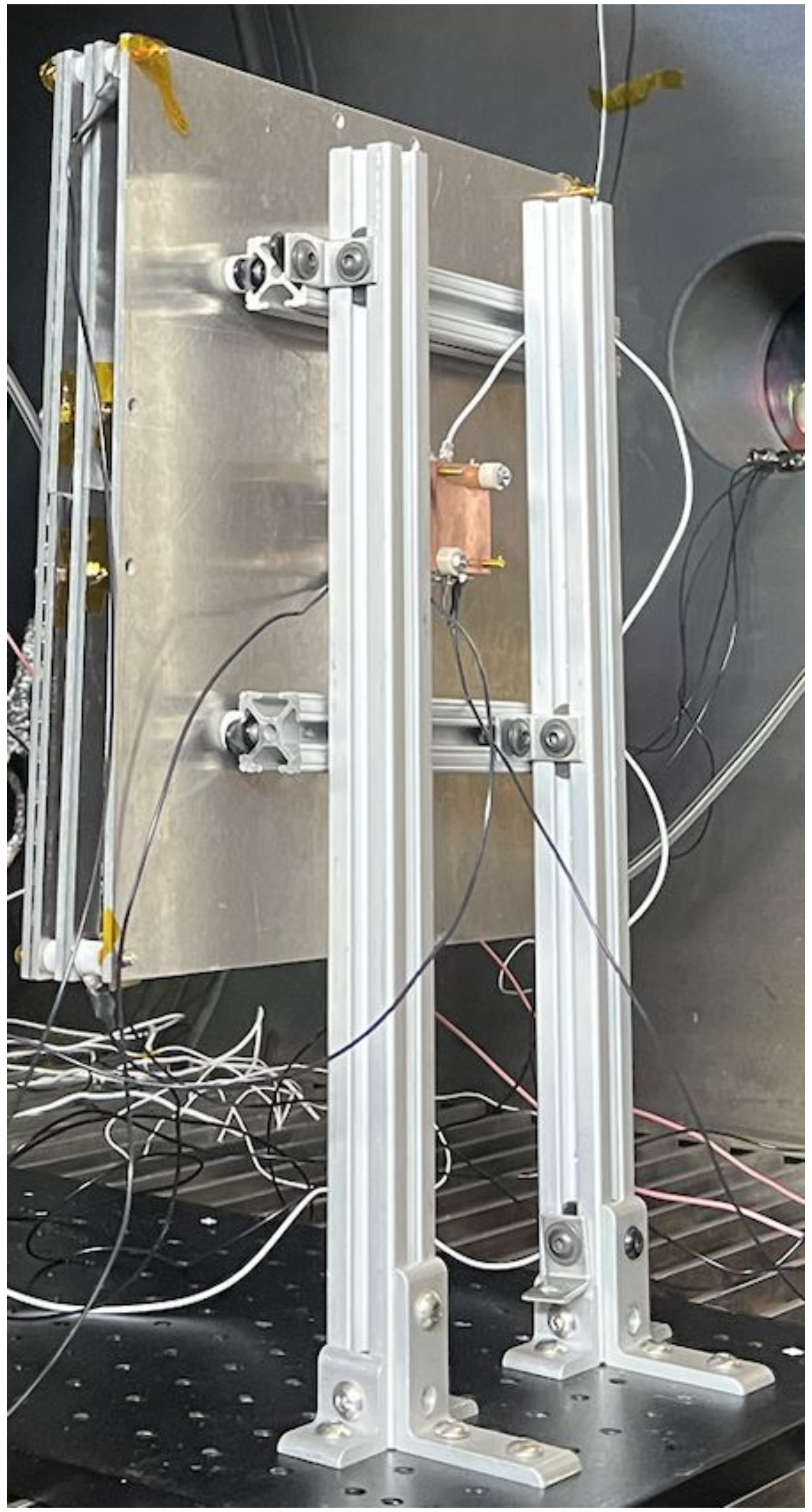
$$\boxed{4\lambda = 5.12\ \mathrm{mm}}$$

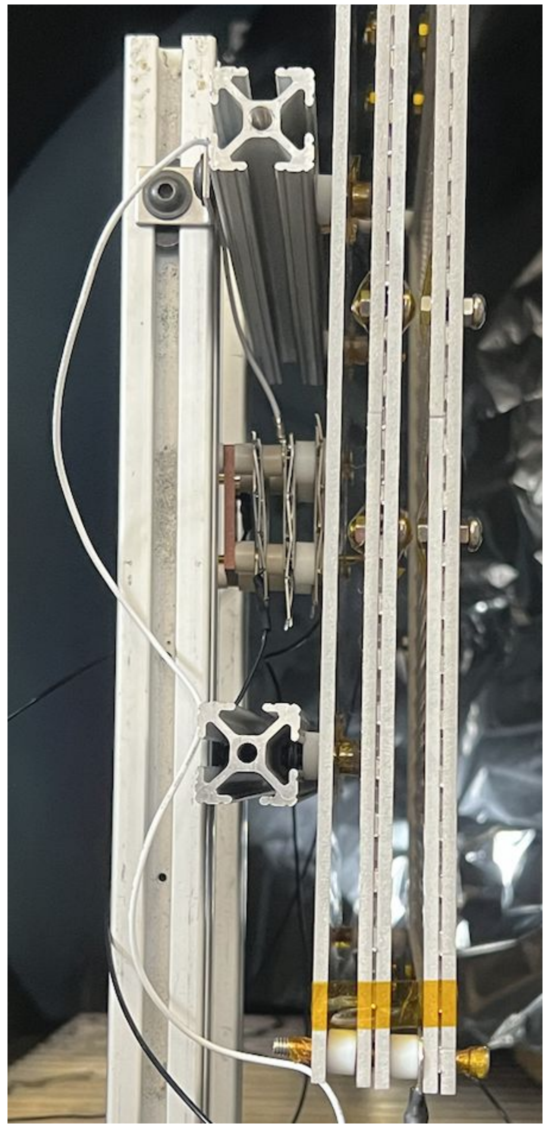
[2] Ferda, B. (2015, September 24). *Retarding potential analyzer theory and design*. Princeton University. https://w3.pppl.gov/ppst/docs/ferda.pdf

Current Collector and RPA Assembly









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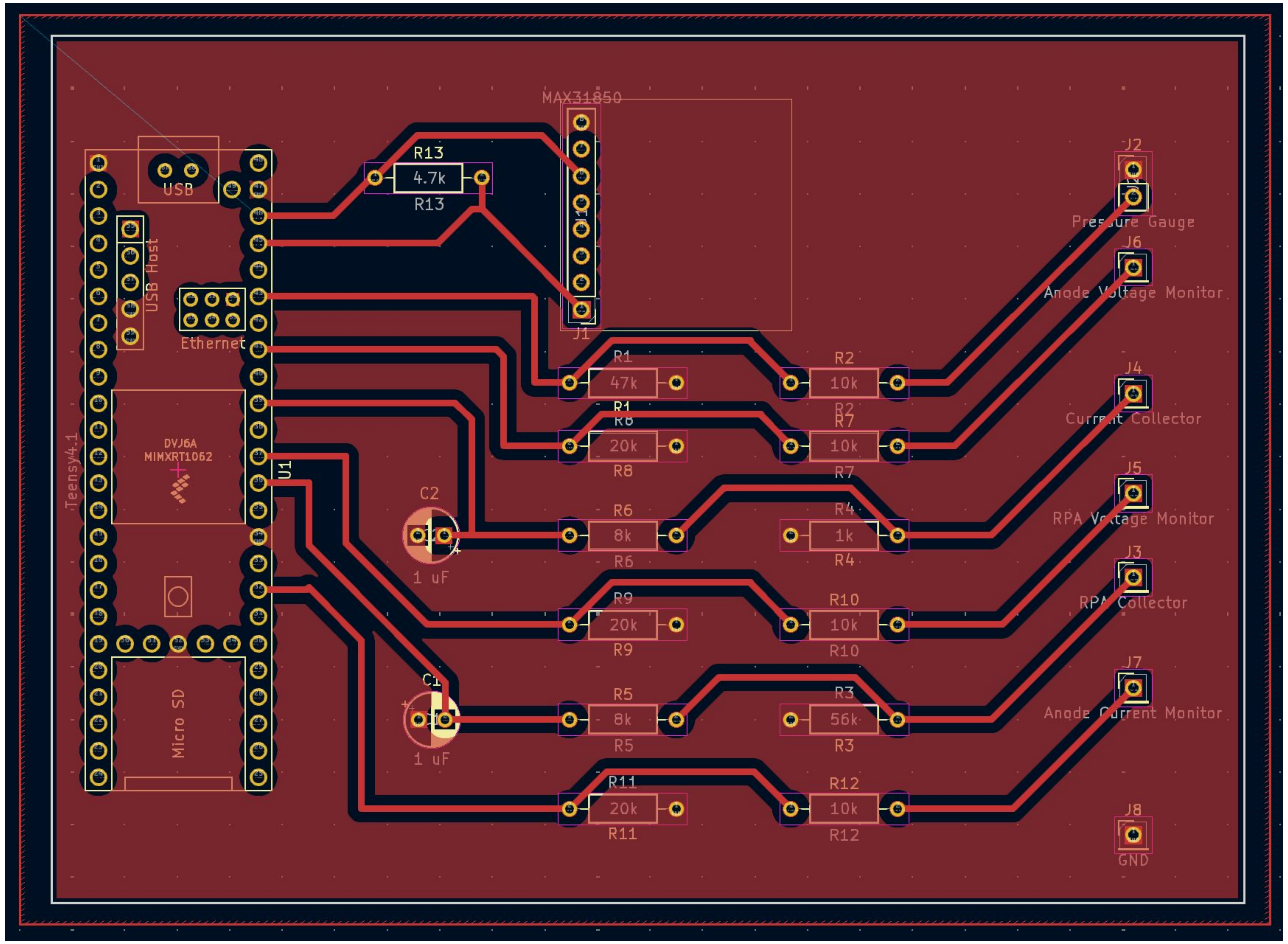
Slide by: EC, Presented by: EL

DAQ


MAX31850
R13
4.7k
R13
J1
USB
USB Host
Ethernet
DVJ6A
MIMXRT1062
Teensy4.1
U1
Micro SD
J2
Pressure Gauge
J6
Anode Voltage Monitor
J4
Current Collector
J5
RPA Voltage Monitor
J3
RPA Collector
J7
Anode Current Monitor
J8
GND
R1
47k
R2
10k
R8
20k
R7
10k
R6
8k
R4
1k
C2
1 uF
R9
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R10
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C1
1 uF
R5
8k
R3
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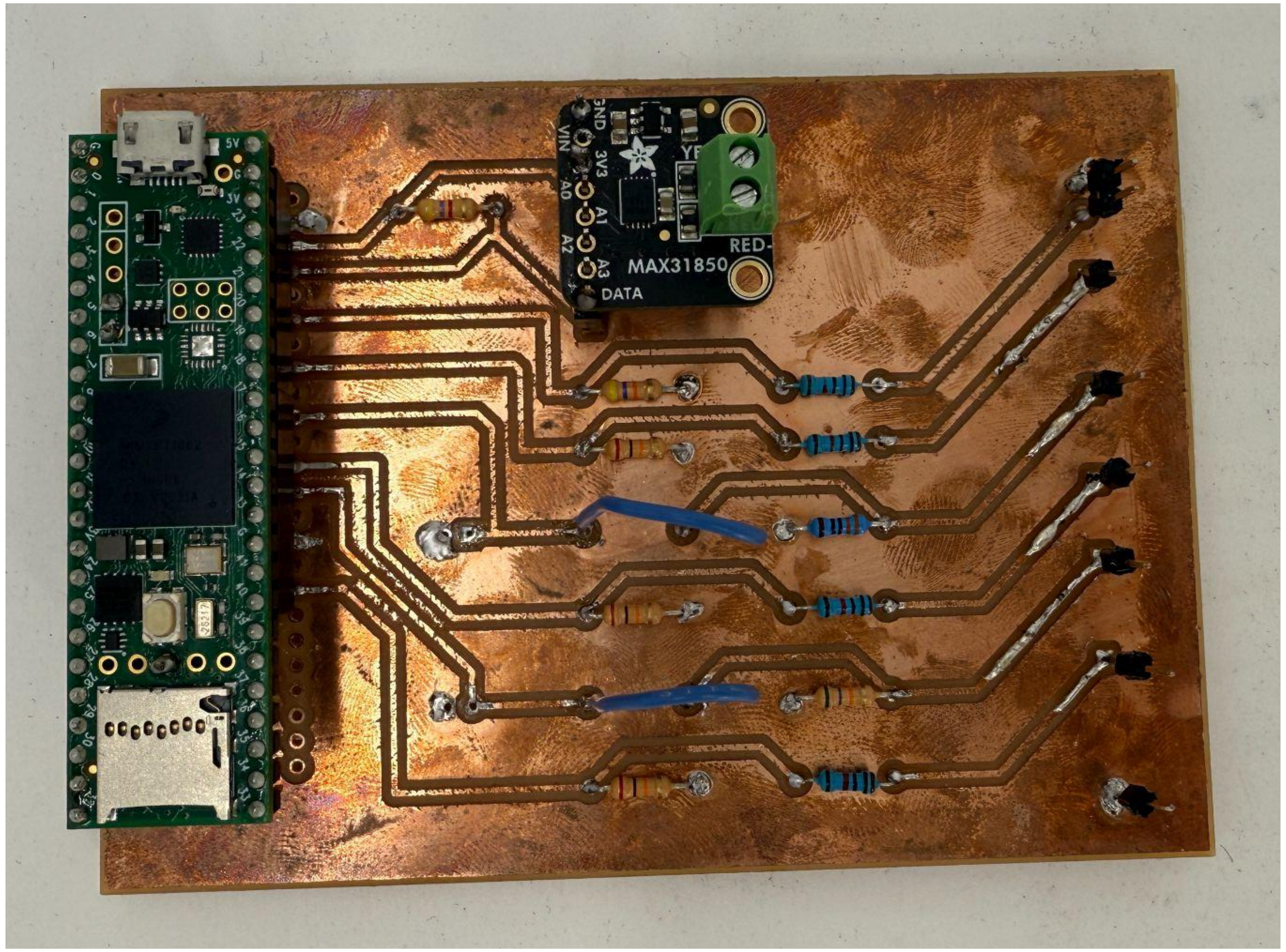
R11
20k
R12
10k


GND
VIN
3V3
A0
A1
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MAX31850
DATA
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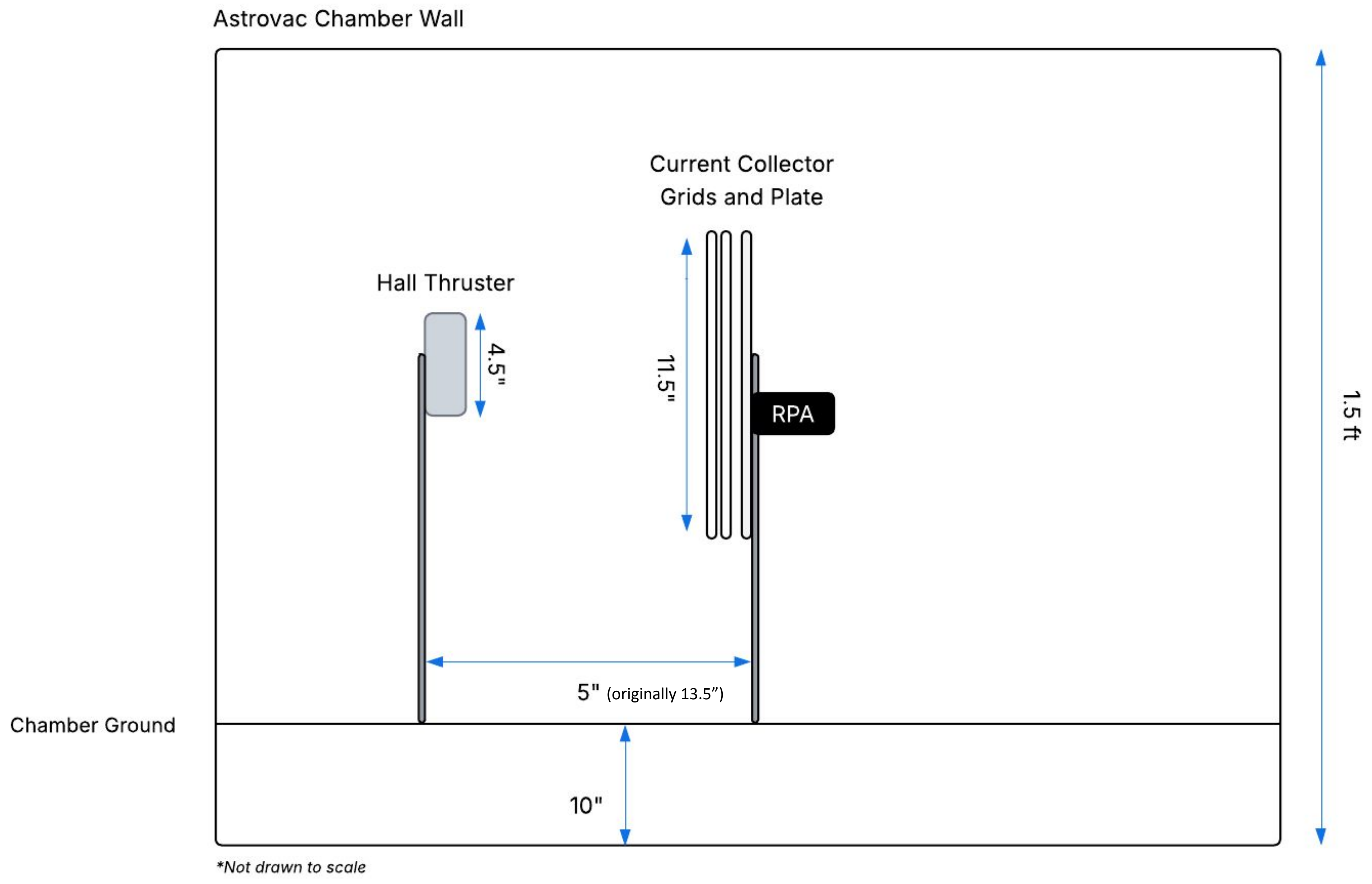
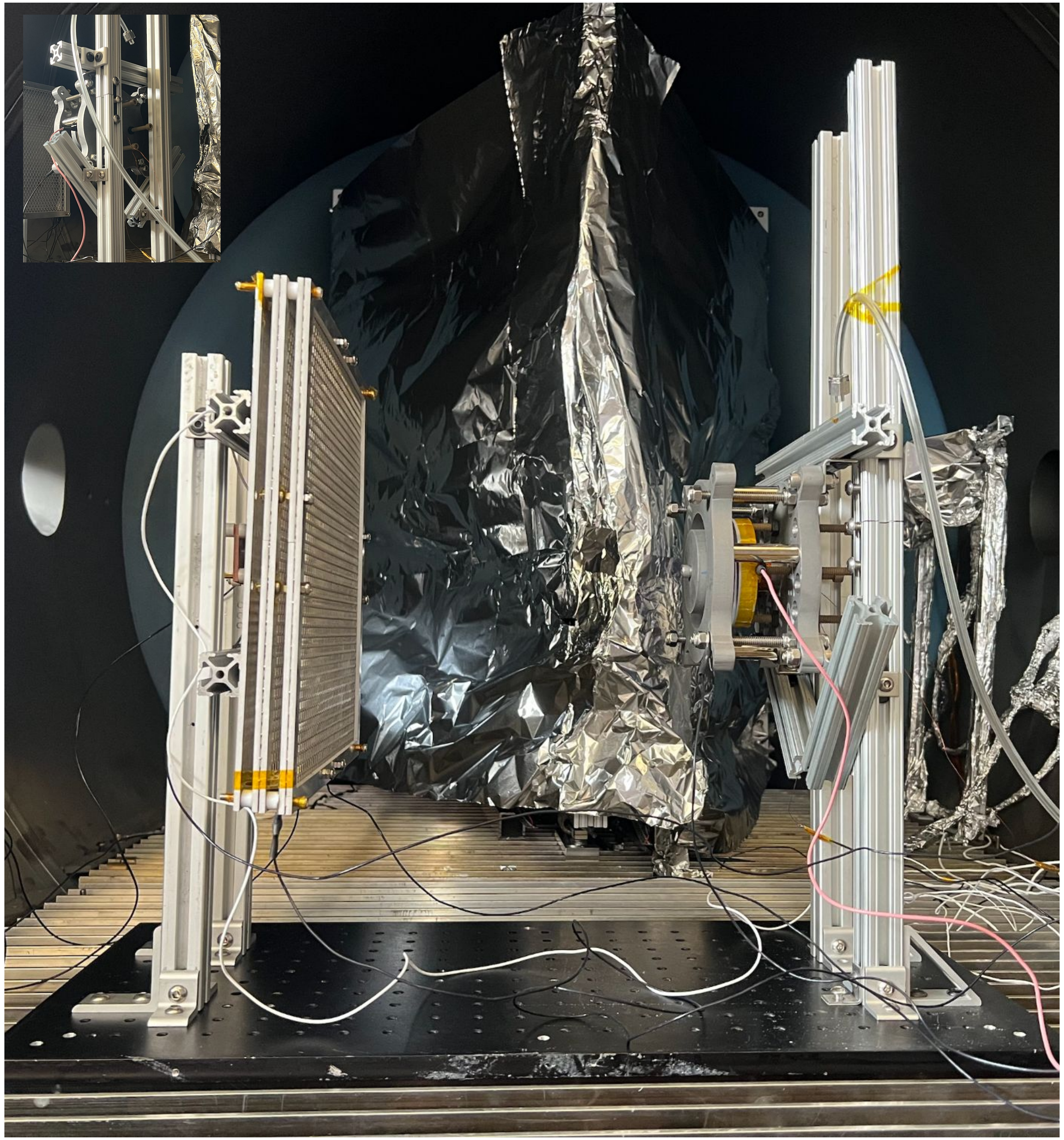
Test Mounts

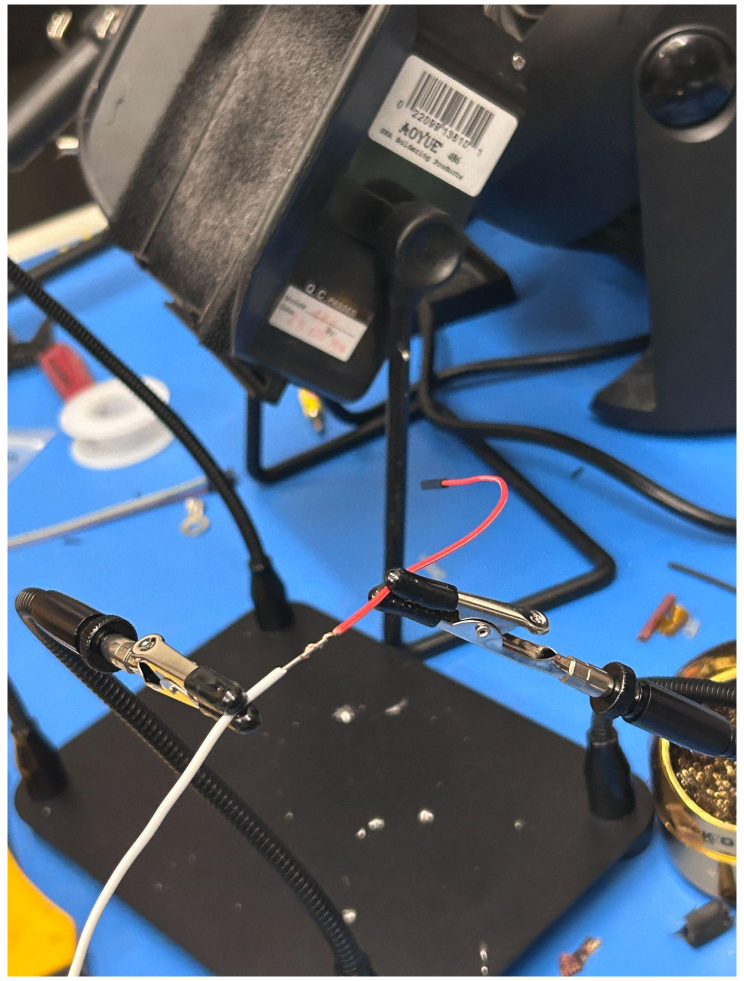




Slide by: NHG & OP, Presented by: NHG & OP

Electrical Connections - Electrical Checkout

#	Termination 1	Destination 1	Termination 2	Destination 2	High voltage?	Inside vacuum?	Wire made/existing?	Wire attached?	Continuity confirmed?	Notes
1	female 1/10" pin connector	DAQ, pressure gauge GND	exposed wire	Chamber wall, pressure gauge GND	No	No	Yes	No	No	need to strip + solder, extend?
2	female 1/10" pin connector	DAQ, pressure gauge signal	exposed wire	Chamber wall, pressure gauge signal	No	No	Yes	No	No	need to strip + solder, extend?
3	female 1/10" pin connector	DAQ, RPA back plate	female D-sub pin connector	Chamber wall, D-sub feedthrough	No	No	Yes	No	No	may need to remake due to heat shrink bein
4	female 1/10" pin connector	DAQ, current collector	female D-sub pin connector	Chamber wall, D-sub feedthrough	No	No	Yes	No	No	may need to remake due to heat shrink bein
5	female 1/10" pin connector	DAQ, RPA voltage monitor	female D-sub pin connector	Table, Bertan 230-05R HV power suppl	No	No	Yes	No	No	may need to remake due to heat shrink bein
6	female 1/10" pin connector	DAQ, anode voltage monitor	female D-sub pin connector	Table, Bertan 205B-10R HV power supp	No	No	Yes	No	No	may need to remake due to heat shrink bein
7	female 1/10" pin connector	DAQ, anode current monitor	female D-sub pin connector	Table, Bertan 205B-10R HV power supp	No	No	Yes	No	No	may need to remake due to heat shrink bein
8	female 1/10" pin connector	DAQ, signal GND	female D-sub pin connector	Table, HV power supplies, pin 7	No	No	Yes	No	No	may need to remake due to heat shrink bein
9	female D-sub pin connector	Chamber wall, D-sub feedthrough	O-ring terminal	Current collector back plate	No	Yes	Yes	No	No	
10	female D-sub pin connector	Chamber wall, D-sub feedthrough	O-ring terminal	RPA, back plate	No	Yes	Yes	No	No	
11	female D-sub pin connector	Chamber wall, BNC feedthrough	O-ring terminal	RPA, mesh 2	No	Yes	Yes	No	No	
12	female D-sub pin connector	Chamber wall, BNC feedthrough	O-ring terminal	RPA, mesh 4	No	Yes	Yes	No	No	
13	female D-sub pin connector	Chamber wall, BNC feedthrough	O-ring terminal	Current collector, mesh 2	No	Yes	Yes	No	No	
14	O-ring terminal	Current collector, mesh 1	O-ring terminal	Chamber, grate	No	Yes	Yes	Yes	No	switched termination 2 from alligator clip to
15	O-ring terminal	RPA, mesh 1	O-ring terminal	Chamber, grate	No	Yes	Yes	No	No	switched termination 2 from alligator clip to
16	BNC	Table, low voltage power supply	BNC	Chamber wall, BNC feedthrough	No	No	Yes	No	No	
17	BNC	Table, low voltage power supply	BNC	Chamber wall, BNC feedthrough	No	No	Yes	No	No	
18	BNC	Table, low voltage power supply	BNC	Chamber wall, BNC feedthrough	No	No	Yes	No	No	
19	MHV	Table, HV power supply	MHV	Chamber wall, HV feedthrough	Yes	No	Yes	No	No	
20	MHV	Table, HV power supply	MHV	Chamber wall, HV feedthrough	Yes	No	Yes	No	No	
21	female spade	Saba's MHV adapter	O-ring terminal	Thruster, anode	Yes	Yes	Yes	No	No	
22	female spade	Saba's MHV adapter	O-ring terminal	RPA, mesh 2	Yes	Yes	Yes	No	No	
23	BNC	Table, low voltage power supply	BNC	Chamber wall, BNC feedthrough	No	No	Yes	No	No	
24	female D-sub pin connector	Chamber wall, BNC feedthrough	???	Magnetic circuit?				No	No	



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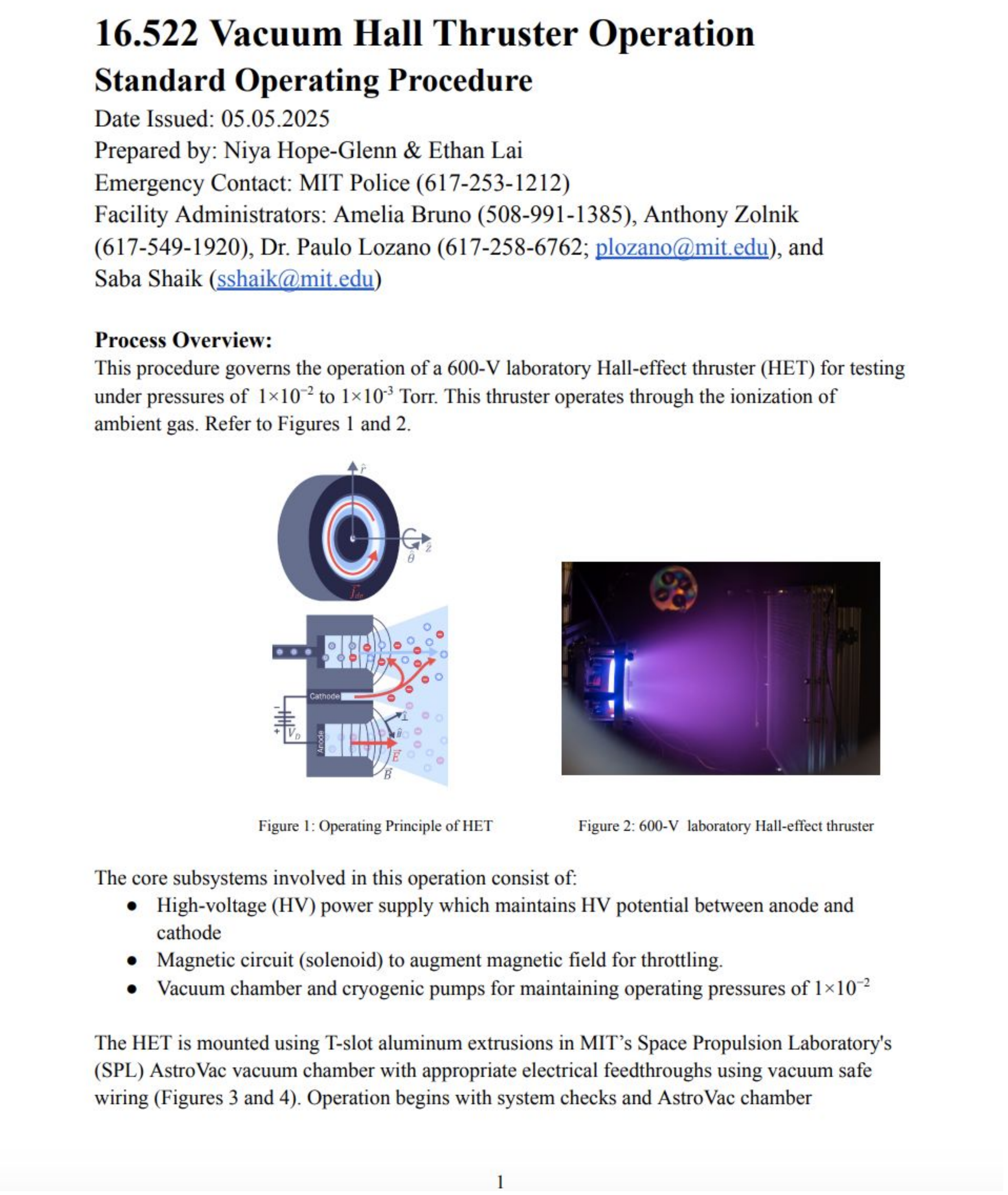
Slide by: ZZ, Presented by: ZZ

User Interface

Supported features:

- **Real time readouts** for
  - Anode monitor voltage
  - Anode monitor current
  - RPA monitor voltage
  - Current collector current
  - RPA current
  - *Pressure gauge reading noted manually (1.25 - 2.71 * 10^-2)*
- **Real time** CC current and RPA current vs time **graphs**
- **Data logging** to csv file

Standard Operating Procedure Document (SOP)



16.522 Vacuum Hall Thruster Operation

Standard Operating Procedure

Date Issued: 05.05.2025

Prepared by: Niya Hope-Glenn & Ethan Lai

Emergency Contact: MIT Police (617-253-1212)

Facility Administrators: Amelia Bruno (508-991-1385), Anthony Zolnik (617-549-1920), Dr. Paulo Lozano (617-258-6762; plozano@mit.edu), and Saba Shaik (sshaik@mit.edu)

Process Overview:

This procedure governs the operation of a 600-V laboratory Hall-effect thruster (HET) for testing under pressures of 1×10^{-2} to 1×10^{-3} Torr. This thruster operates through the ionization of ambient gas. Refer to Figures 1 and 2.

Figure 1: Operating Principle of HET

Figure 2: 600-V laboratory Hall-effect thruster

The core subsystems involved in this operation consist of:

- High-voltage (HV) power supply which maintains HV potential between anode and cathode
- Magnetic circuit (solenoid) to augment magnetic field for throttling.
- Vacuum chamber and cryogenic pumps for maintaining operating pressures of 1×10^{-2}

The HET is mounted using T-slot aluminum extrusions in MIT's Space Propulsion Laboratory's (SPL) AstroVac vacuum chamber with appropriate electrical feedthroughs using vacuum safe wiring (Figures 3 and 4). Operation begins with system checks and AstroVac chamber

1

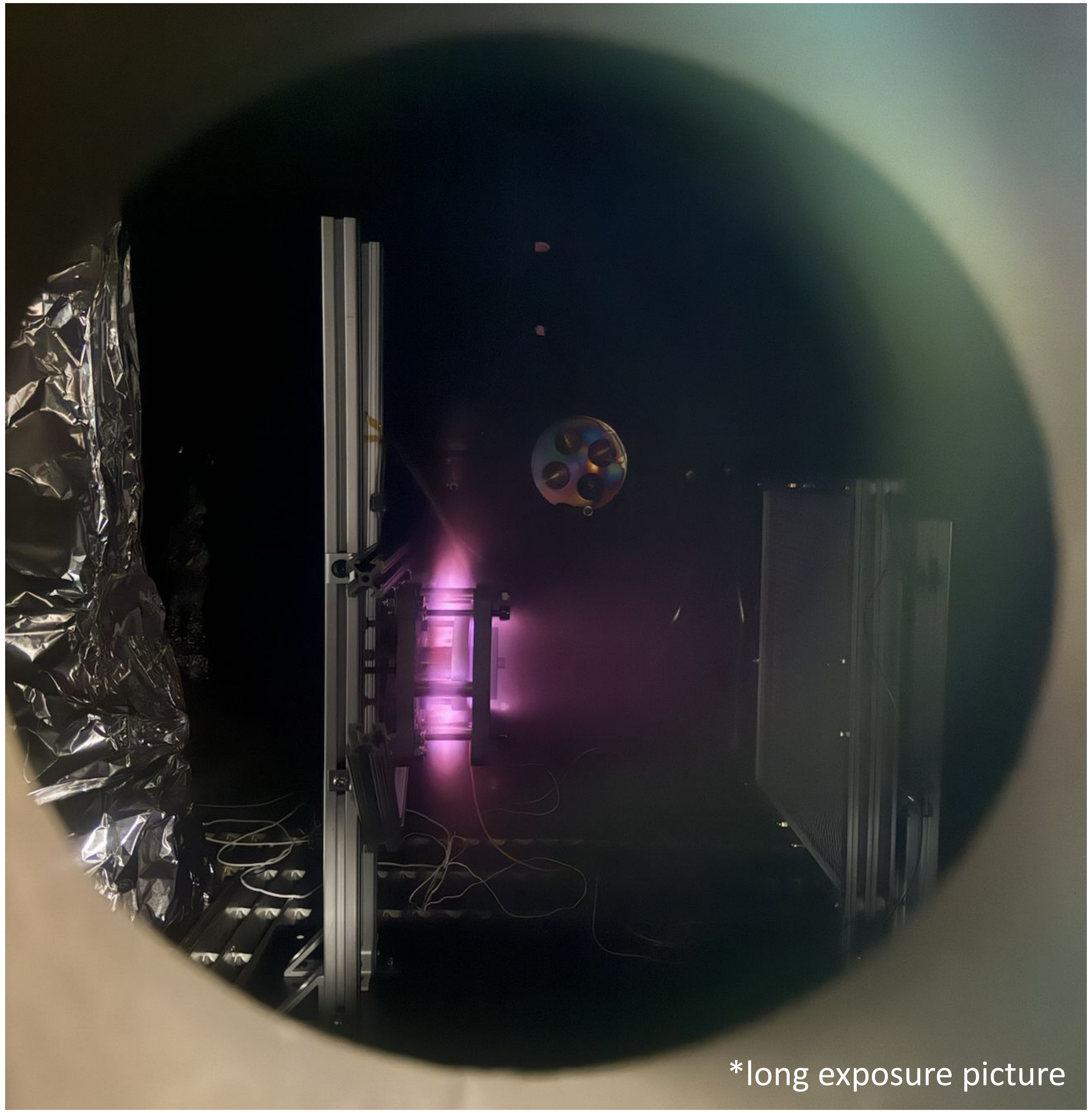
- Creation of 13-page Standard Operating Procedure document to ensure safety of testing operators and equipment
  - Ensures that testing operators complete continuity checks and ensure thruster grounding prior to testing
  - Detailed emergency shutdown procedure in the event of an emergency is clearly defined
- Real time monitoring requirements (off-nominal conditions and troubleshooting suggestions listed)
- Permits safe operation for new operators

*Refer to Appendix 7 for full SOP document.

Testing Results

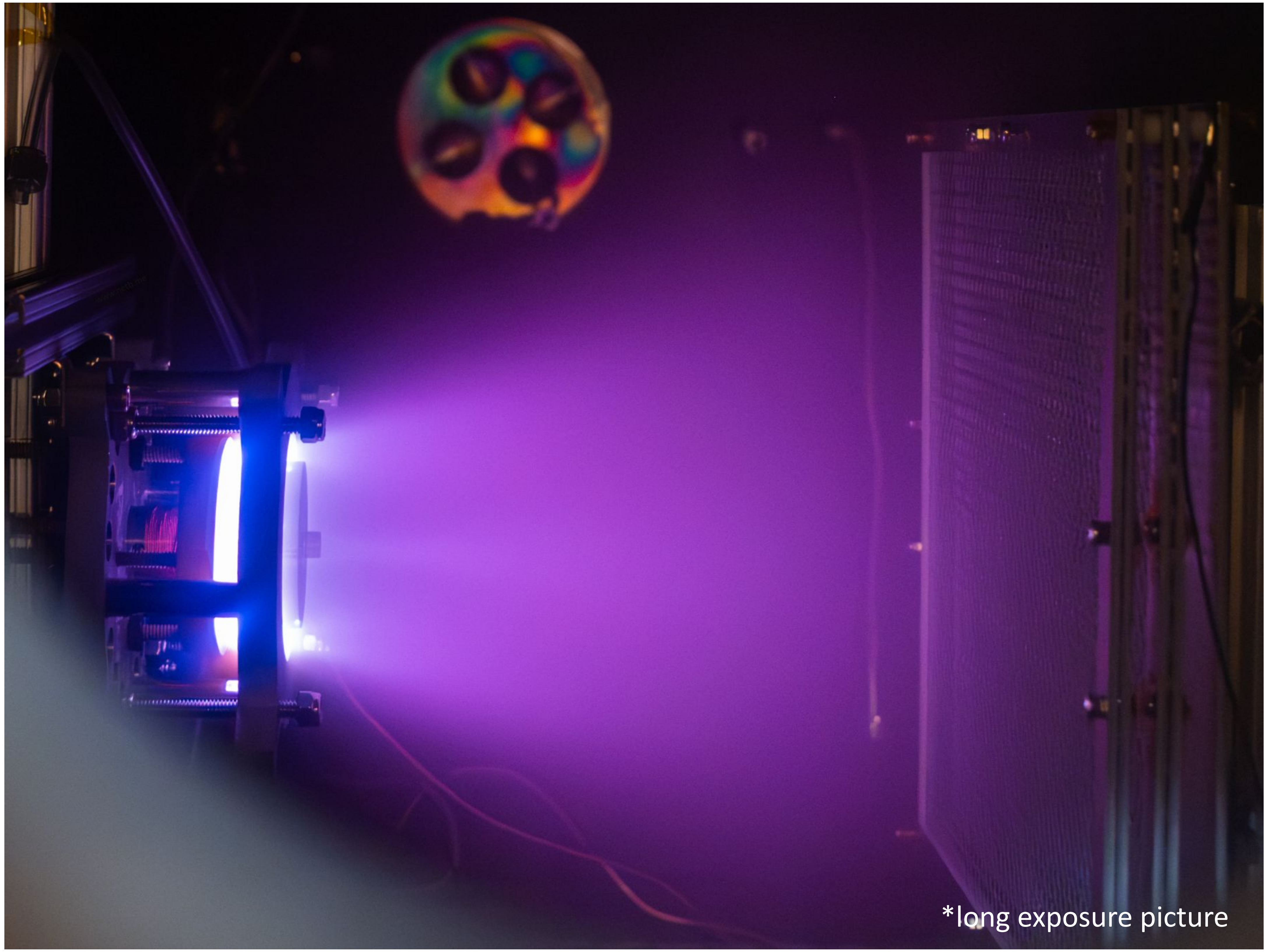
Sub-Teams 5 & 6

Rough Vacuum Test 1 (5/5)



*long exposure picture

Rough Vacuum Test 2 (5/6)



- More directed plasma flow towards RPA
- Radial sparking on thruster edges
- Did not collect accurate data

For the next test:

- Kapton tape on protruding edges or sharp junctions to avoid arc discharge points
- Sanding thruster geometry to eliminate protruding edges from created localized fields
- Completed SOP prior to future tests- permits new test operators to follow a standardized procedure for hooking up electrical connections (refer to Slide 42)

Rough Vacuum Test 3 (5/7)

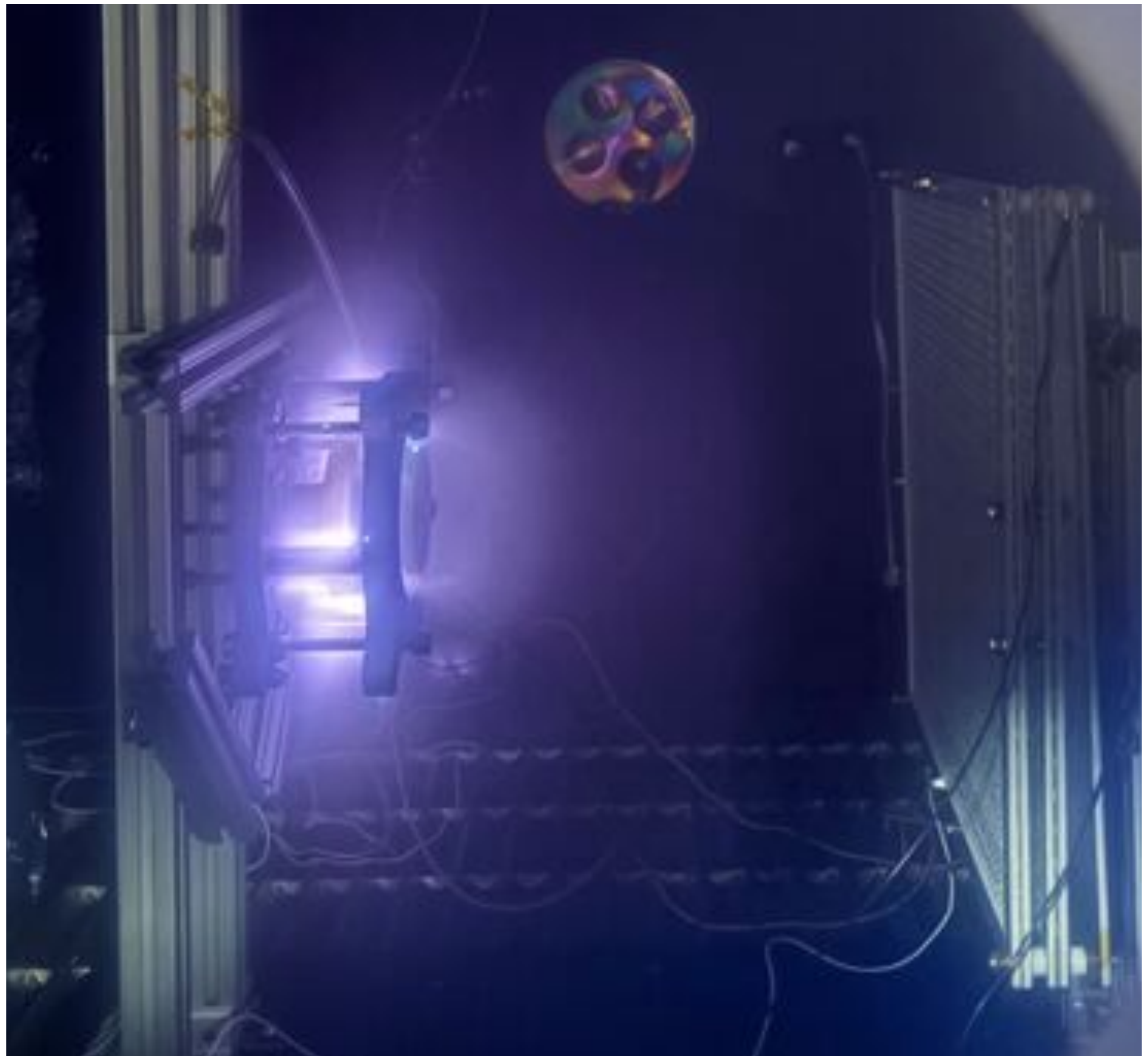
What happened?

- plume is still very faint, glowing behind the thruster
- sparking a lot again

Next steps

- retape the thruster
- double check all connections
- switch out power supply (for higher current)

Rough Vacuum Test 4 (5/9)



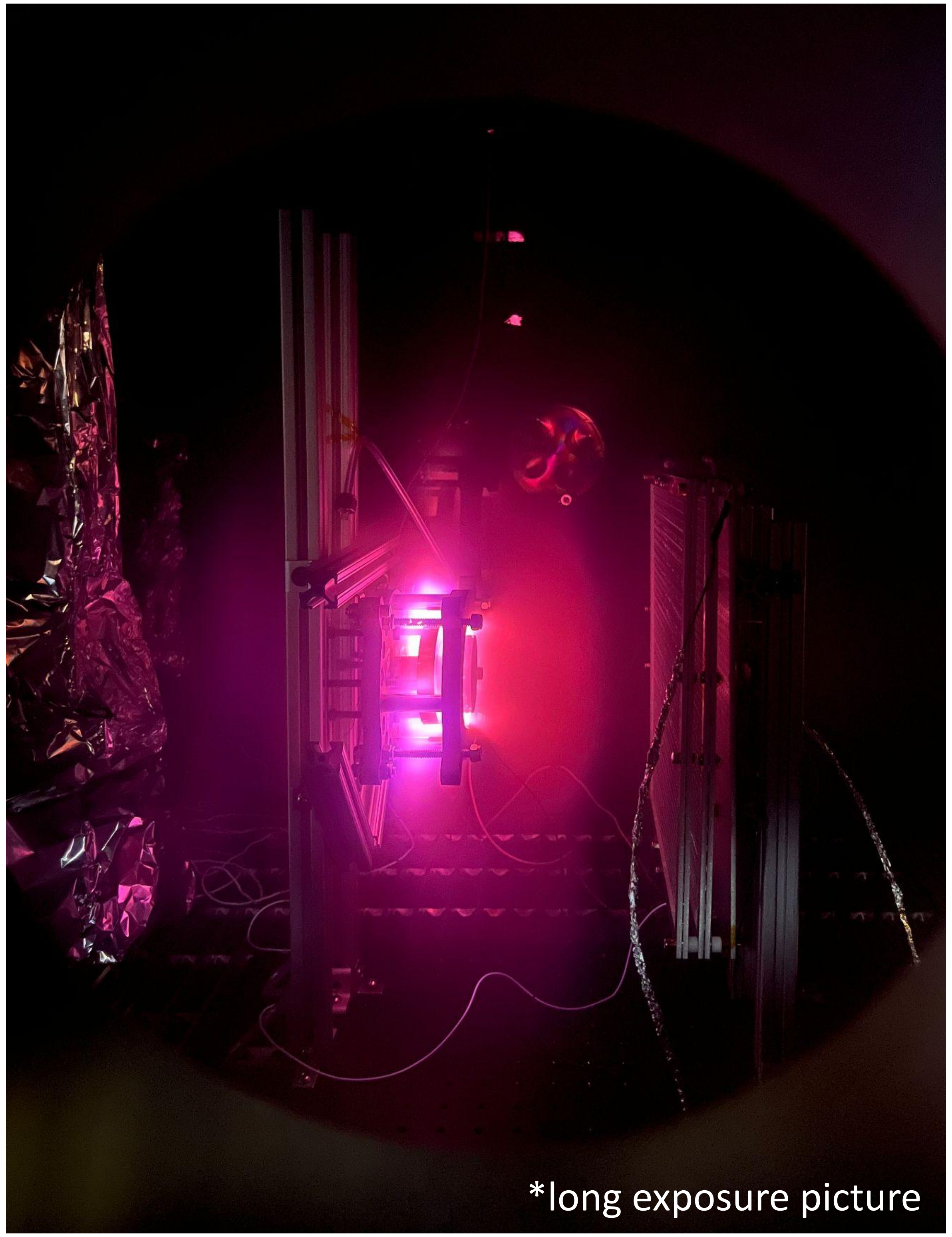
(1.25 * 10 ^-2 torr)

- **What went wrong?**
  - Small plume
  - Glowing behind anode
- **Why?**
  - Degraded insulating tape
  - Operating at higher current than test 3, switch back to original power supply

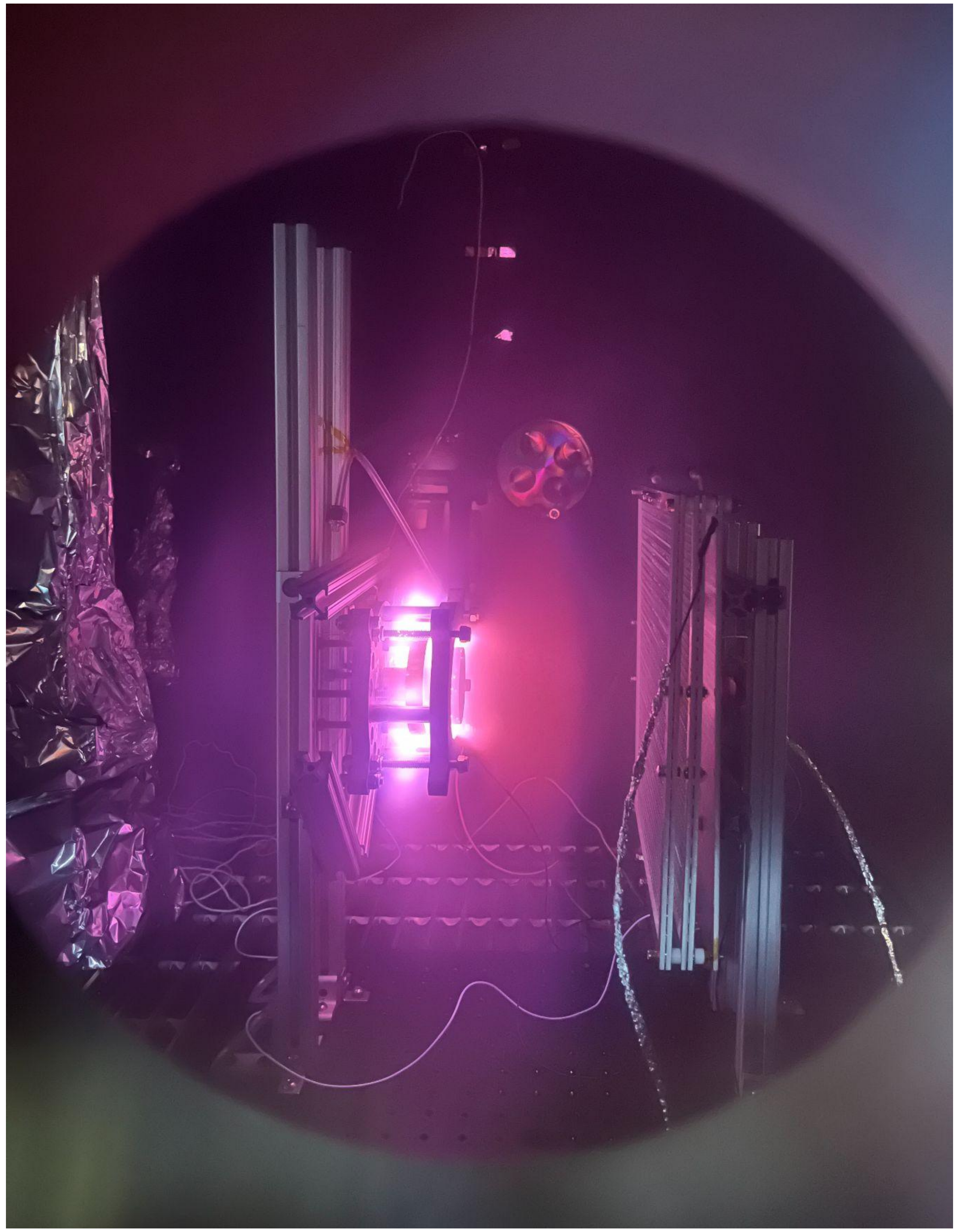
Rough Vac Test 4

- plume keeps getting fainter??
- move the current collector closer
- re-do the tape
- check electrical connections / grounding
- shield all current readout wires with foil
- fix daq calibration issues

(Last) Rough Vacuum Test 5 (5/12)



(2.75 * 10 ^-2 torr)



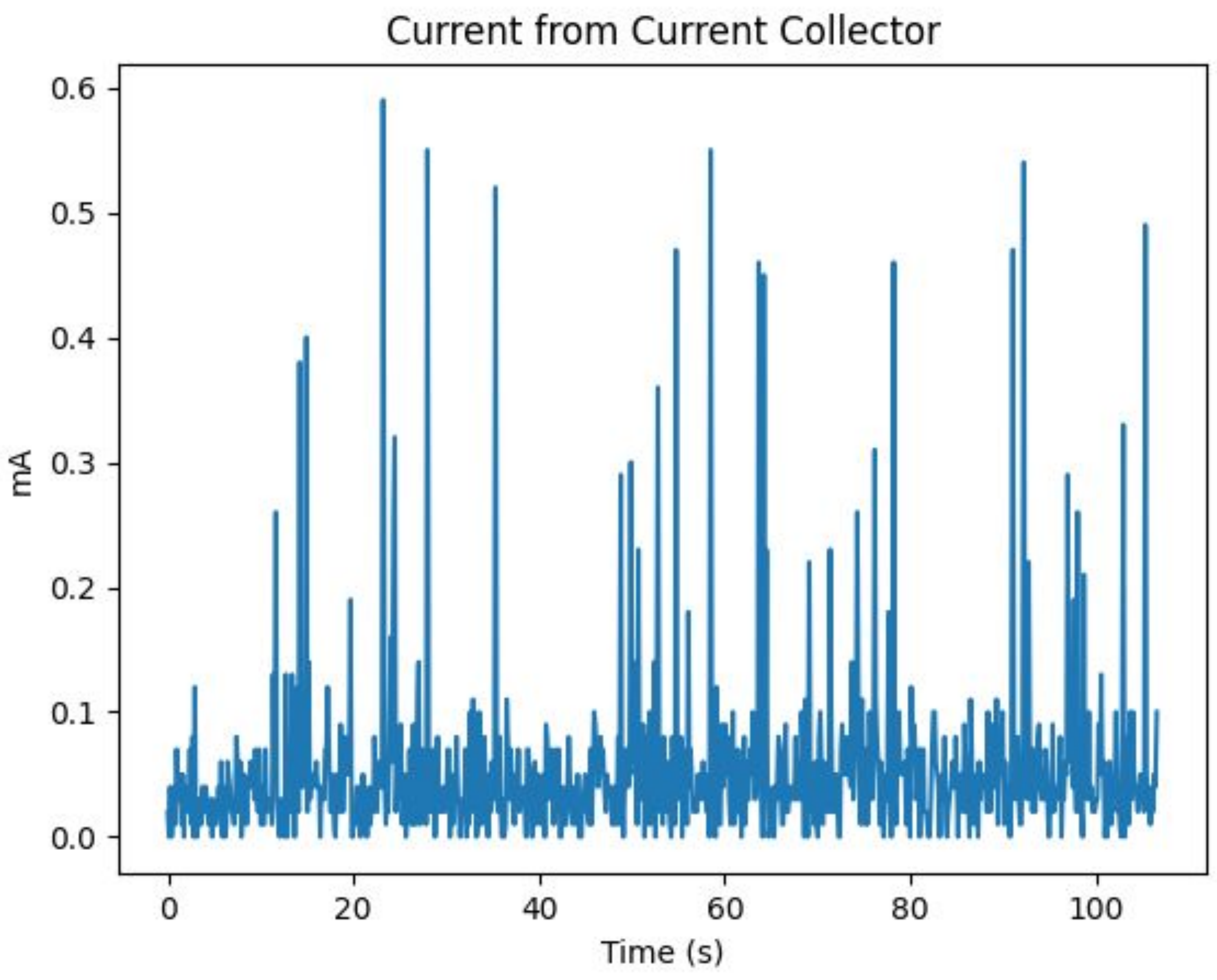
Final round of testing!

- moved current collector closer to the thruster
  - from 13.5” to 5”
- foil around wires and calibrated calculations for better data acquisition
- properly grounded DAQ
- plume is much brighter!
- plume is still glowing behind the anode but less so
- some funky data… but still a successful test!

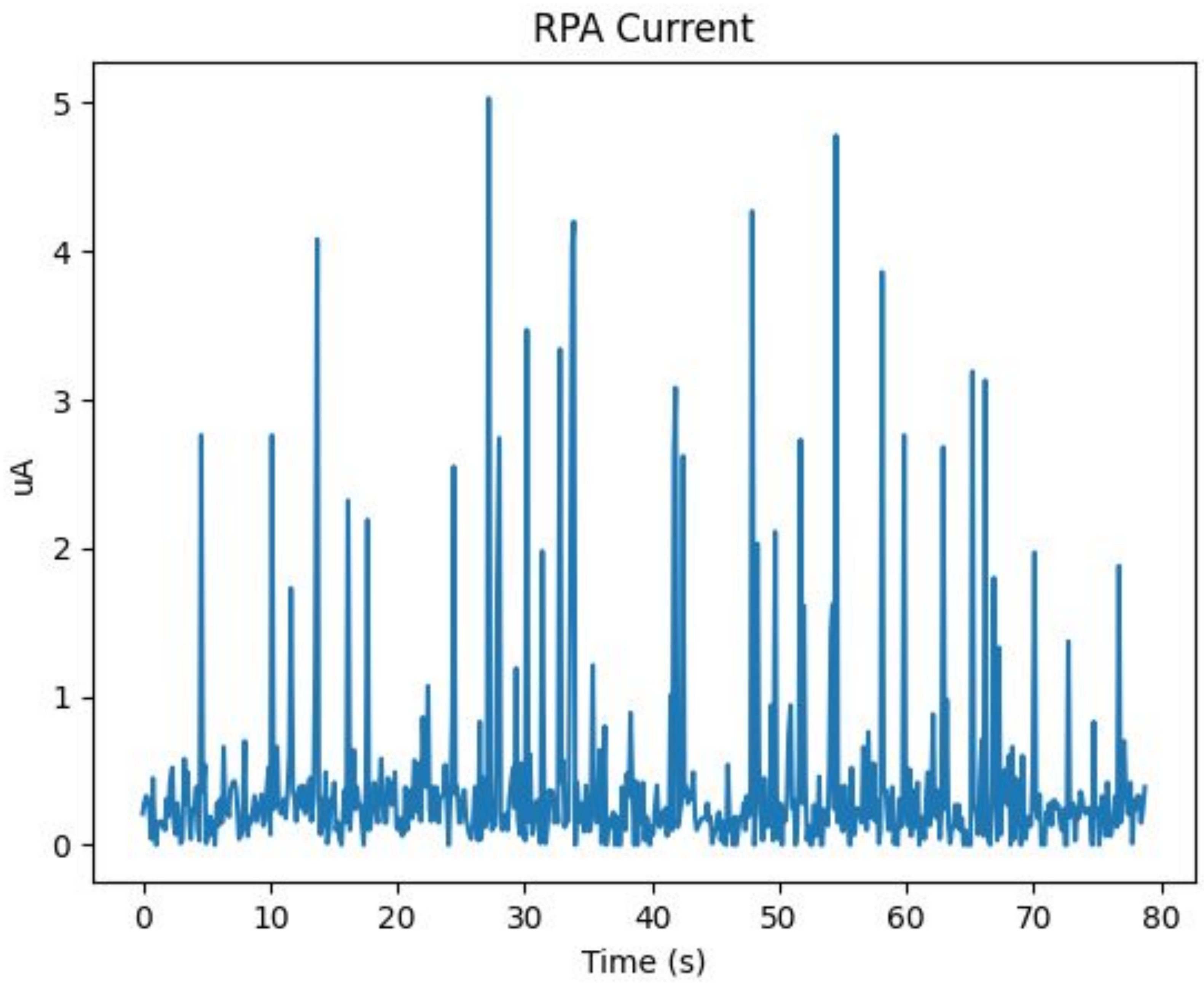
Raw Results

Anode operation:
- **2.5 mA**
- **480 V**

Current Collector



RPA



Expected Results

Current Collector

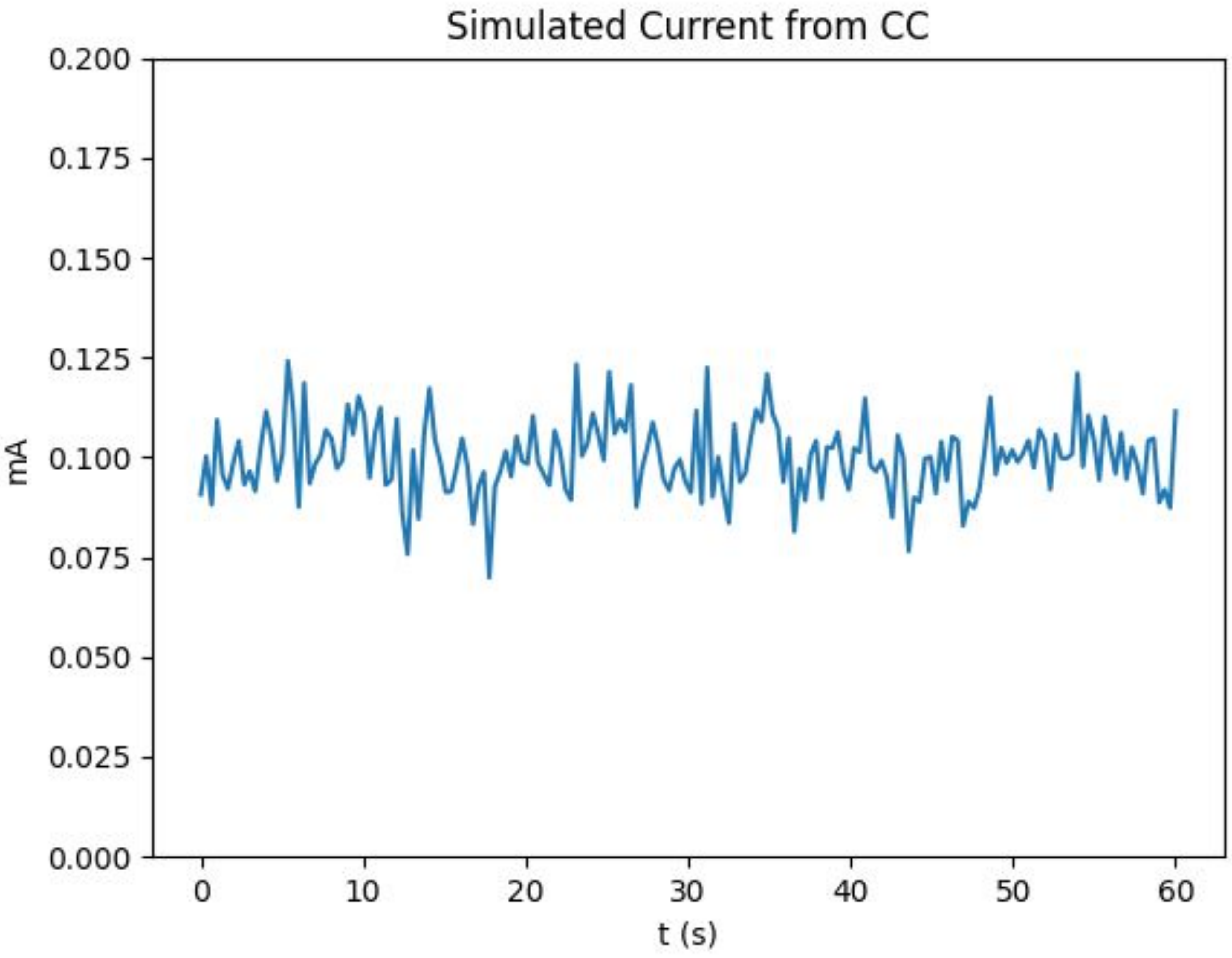
$$\dot{m} = I_{CC}\frac{m}{q}$$

$$\eta_a = \frac{I_{CC}}{\phi_t I_a}$$

- Assume singly charged N2 molecules
- Mesh transparency of 81.5%

$$\dot{m} \approx 3.57 \times 10^{-11}\ \mathrm{kg/s}$$

$$\eta_a \approx 0.05$$



Expected Results

RPA

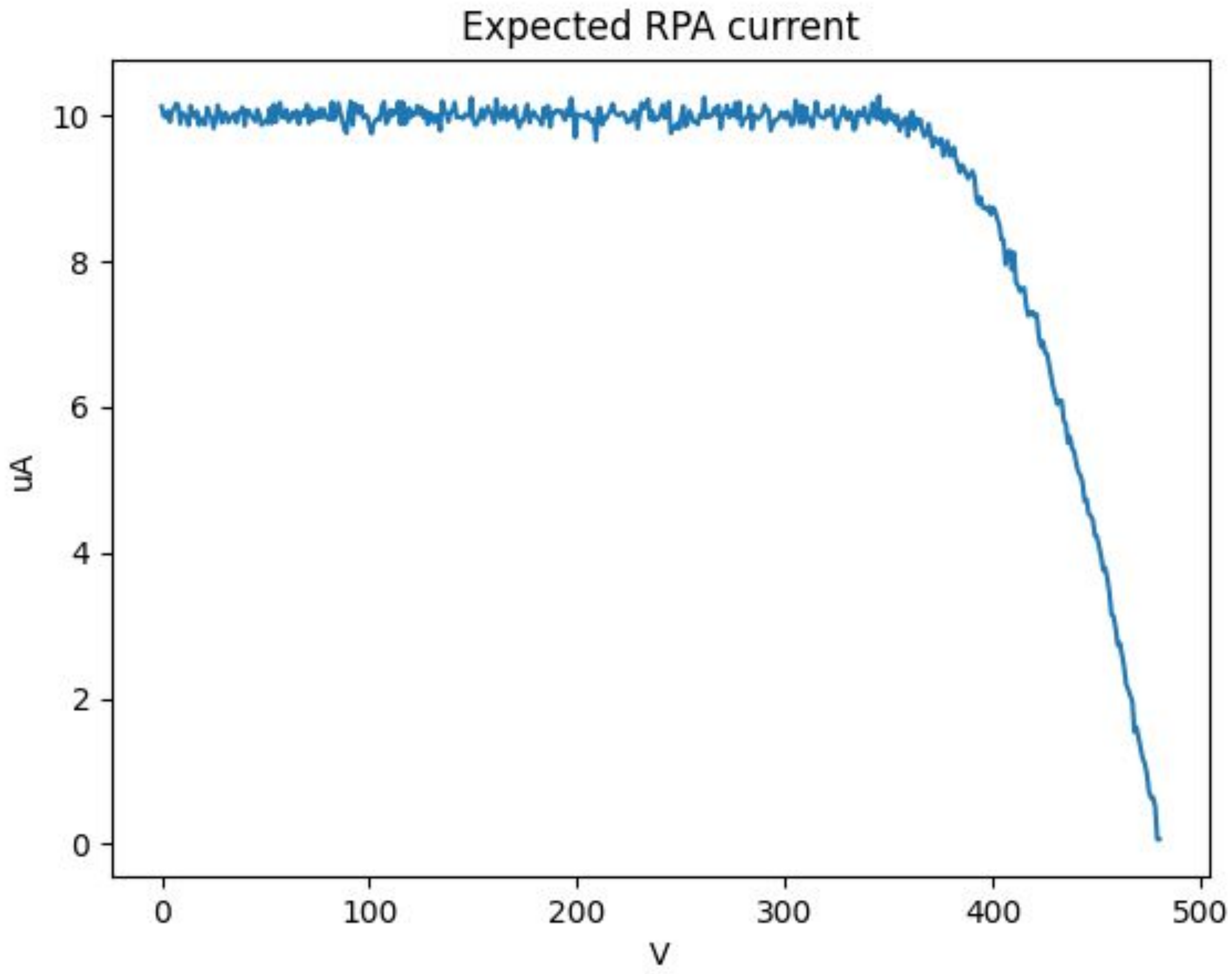
$$\bar{c} = \sqrt{2\frac{q}{m}V_{\mathrm{rep}}}$$

$$F = \dot{m}c,\ \eta = \frac{Fc}{2\mathbb{P}}$$

$$\eta_u \approx 1,\ \eta_e = \frac{\eta}{\eta_a \eta_u}$$

$$F = 1.75 \times 10^{-6}\ \mathrm{N},\ I_{sp} = 5002\ \mathrm{s}$$

$$\eta = 0.036, \eta_e = 0.73$$

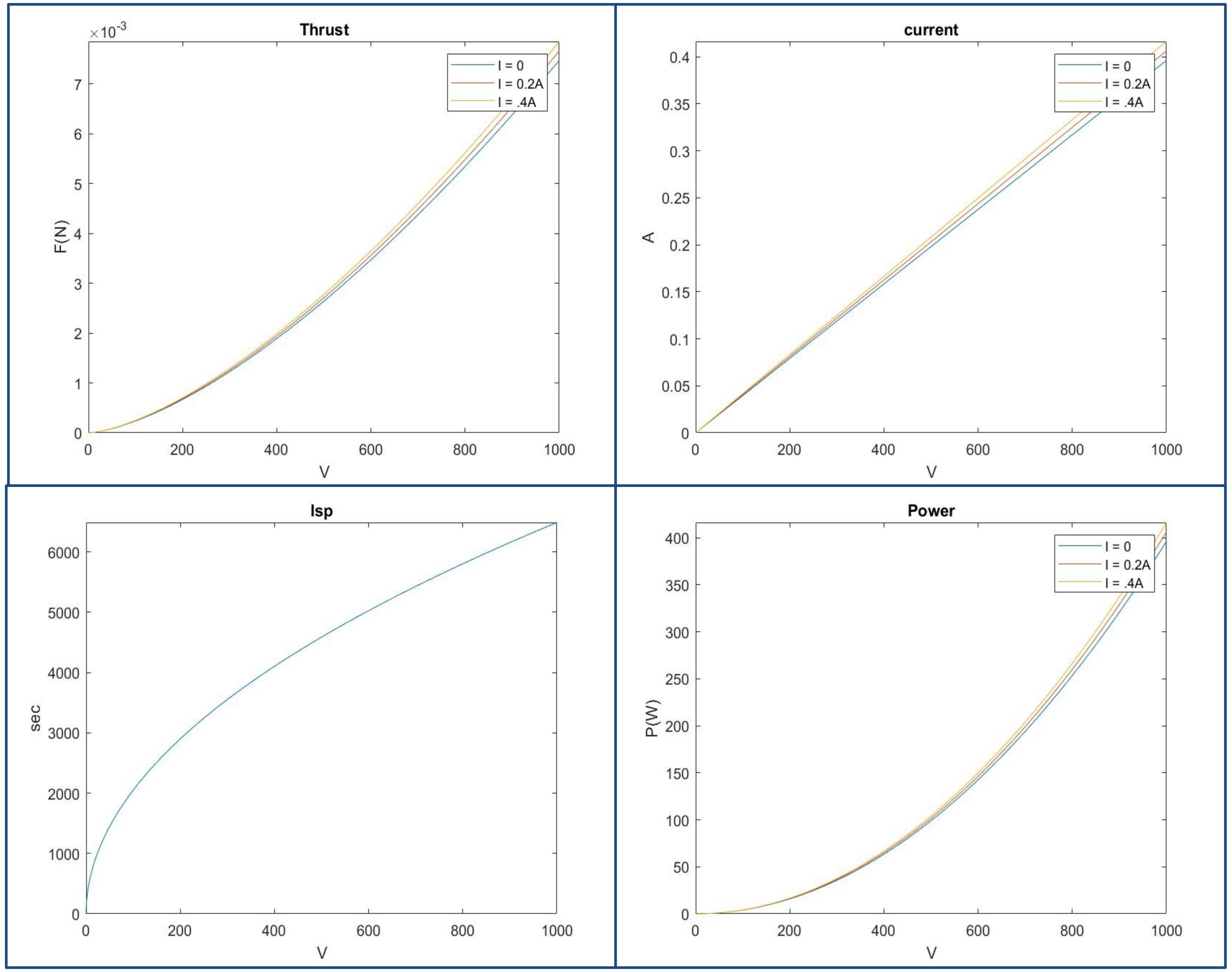


Conclusion and Future Work

TRL Evaluation

Semi-Empirical Evaluation

- Academic and industrial hall thrusters [1,2]:
  - Voltage: 200 V
  - Thrust: 2-7 mN
  - ISP: 1000s
  - Power: 100W
- Our empirical calculations using measured magnetic fields are loosely aligned with academic and industrial hall thrusters



TRL Evaluation

Potential Application

- Stationkeeping for LEO and vLEO satellites
- Unable to test at LEO pressures
  - Rough Vac (~.02 torr) corresponds to 10-20 km above sea level

Evaluation

- TRL 2
  - Application is speculative
  - No detailed analysis to support analytical predictions
  - Basic properties of algorithms, representations and concepts defined
- Can level up to TRL 3 by producing clean data collection
- Can further level up to TRL 4 by testing at lower pressures

Conclusions

The team successfully developed a VHT with a magnetic field vernier controller and completed multiple firings. Based on the analysis and obtained data, **the team assessed the TRL of the system as 2**, considering a potential application with station keeping in a LEO environment.

No.	Project Objectives	Evaluation	Rationale
1	Design, build and test a VHT with Magnetic Field Vernier control	**Good**	• **A VHT was successfully developed** using four permanent magnets and a 100-turn copper coil, **guided by 2D Maxwell simulations and unit testing**. • **The thruster was successfully fired with magnetic field vernier control.**
2	Design, build and use an RPA to characterize your VHT	Partial	• **An RPA was developed** with 5.12 mm grid spacing configuration, based on Debye length. • However, **the RPA couldn't characterize the thruster's performance** during testing.
3	Propose a theoretical framework to predict the performance and justify your design	**Good**	• **A theoretical framework was developed** incorporating thrust, Isp, power, current, and magnetic field requirement **and justified the VHT design**.
4	Measure relevant quantities to establish thruster performance	Partial	• **A measurement setup was established**, enabling successful data acquisition to evaluate thruster performance. However, **the experimental data couldn't validate the thruster performance** predicted by the analysis.
5	Postulate and justify the TRL of your VHT design at end-of-effort	**Good**	• Based on testing and analysis, **the VHT was assessed to meet TRL 2 criteria**, with a prospective application **with station keeping in a LEO mission**.

Lessons Learned

No.	Team	Lessons Learned
1	Project Management	● Conduct a more detailed budget analysis and don’t to be too concerned on shipping time ● Allocate more time for assembly/mounting (only tested for < 1 week)
2	Magnetic Circuit	● Theoretical sizing overestimates the magnetic field by 5.5~9.3 times likely due to no magnetic leak consideration. We should have prioritized Maxwell simulation or unit testing. *Even Maxwell 2D simulation overestimates the magnetic field by 1.5 ~ 2 times. ● Couldn’t conduct full 3D maxwell simulation due to the number of mesh constraints. We should’ve tried with a simpler configuration like 2D computation from the beginning.
3	RPA	● Select materials for easier machining
4	Thruster	● Measure twice, or three times, cut once
5	Testing facilities	● Check connections! ● SOP & scheduling should be completed prior to first test
6	Protocols and Analysis	● Experimental characterization of DAQ ● Space-charge saturation

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Acknowledgements

We gratefully acknowledge the following individuals who have supported us throughout the design and construction process:

Prof. Paulo Lozano
Saba Shaik
Todd

Thank You!

Any Questions?

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7. Paulo Lozano, Lecture 16, April 8

Conclusion and TRL References

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Appendix

Appendices 1-5: Project Contributions
Appendix 6: TRL Definitions
Appendix 7: External Project References

Appendix 1 - Sub-Team 1 Individual Contributions

Niya Hope-Glenn: Created shared Google Drive for subteams to ensure thorough progress documentation, compiled subteam feedback into a working Project Timeline, created and organized final project presentation, helped run team-lead meetings, and tracked/coordinated action items, organized TRL discussions

Mia Tian: Created budget, organized shipping and collection throughout project, organized final project presentation, took running notes for weekly PM meetings, helped run team-lead meetings, tracked/coordinated action items, organized TRL discussions

Daiki Terakado: Supported project management activities, developed the initial outline and helped organize the final project presentation, helped run team lead-meetings, helped organize TRL discussions

Appendix 2 - Sub-Team 2 Individual Contributions

Jordan Bergmann: Researched anode position configurations, set up and computed 2D Maxwell simulations, created simulation figures and graphed results, assisted with magnetic field measurements

Edgar Batista: Researched thruster and anode position configurations, helped set anode position, conducted magnetic field calculations, assisted with magnetic field measurements

Tomas Cantu: Constructed the magnetic circuit, set magnetic field requirements, conducted magnetic field measurements, integrated the circuit into the test configuration.

Daiki Terakado: Set magnetic field requirements, conducted theoretical magnetic field sizing, constructed magnetic circuit, conducted magnetic field measurements, helped set anode position, facilitated the sub-team meetings

Appendix 3 - Sub-Team 3 Individual Contributions

Eddy - Researched RPA theory, helped designed and CAD RPA, Machining parts (waterjet and lathe), Helped assemble and attach RPA to CC

Ethan - Helped design RPA, Machined parts (waterjet and lathe), Helped assemble and attach RPA to CC and make sure wirings are correct

Braeden - Helped design and CAD RPA, Manufactured grids, Helped assembled and attach RPA to CC

Appendix 4 - Sub-Team 4 Individual Contributions

Chris: Design, Machining, Assembly of HET

Braeden : Helped with the design and CAD process, machined the magnetic core, and developed thrust predictions across tested voltage and magnetic values

Ezra: Machining, Material Selection, Current Collector CAD

Appendix 5 - Sub-Teams 5 & 6 Individual Contributions

Nigel: Test stand manufacturing & integration help, that one picture of test 2

Edgar Batista: Test stand manufacturing & integration

Ethan Lai: DAQ design and manufacturing, current collector design and manufacturing, RPA design and manufacturing, writing SOP, testing operations, data analysis, miscellaneous testing tasks

Ottavia Personeni: GUI design and code, Teensy script updates, testing mounts/connections/procedure work, testing operations, data analysis

Niya Hope-Glenn: Assisted with the assembly of testing mounts, wrote SOP, assisted with wiring and electrical connections, created electrical testing schematics, and assisted with miscellaneous testing tasks

Mia Tian: Mounting, electrical connections/wiring

Appendix 6 - NASA Technology Readiness Levels (TRL) Definitions

NASA TECHNOLOGY READINESS LEVELS

From (1) NASA Procedural Requirements 7120.8, Appendix J and (2) NASA Procedural Requirements 7123.1A, Table G.19

TRL	Definition	Hardware Description	Software Description	Exit Criteria
1. Basic principles observed and reported.	Lowest level of technology readiness. Scientific research begins to be translated into applied research and development. Examples might include paper studies of a technology's basic properties.	Scientific knowledge generated underpinning hardware technology concepts/applications.	Scientific knowledge generated underpinning basic properties of software architecture and mathematical formulation.	Peer reviewed publication of research underlying the proposed concept/ application.
2. Technology concept and/or application formulated.	Invention begins. Once basic principles are observed, practical applications can be invented. The application is speculative, and there is no proof or detailed analysis to support the assumption. Examples are still limited to paper studies.	Invention begins, practical application is identified but is speculative, no experimental proof or detailed analysis is available to support the conjecture.	Practical application is identified but is speculative, no experimental proof or detailed analysis is available to support the conjecture. Basic properties of algorithms, representations and concepts defined. Basic principles coded. Experiments performed with synthetic data.	Documented description of the application/concept that addresses feasibility and benefit.
3. Analytical and experimental critical function and/or characteristic proof of concept.	At this step in the maturation process, active research and development (R&D) is initiated. This must include both analytical studies to set the technology into an appropriate context and laboratory-based studies to physically validate that the analytical predictions are correct. These studies and experiments should constitute "proof-of-concept" validation of the applications/concepts formulated at TRL 2.	Analytical studies place the technology in an appropriate context and laboratory demonstrations, modeling and simulation validate analytical prediction.	Development of limited functionality to validate critical properties and predictions using non-integrated software components.	Documented analytical/experimental results validating predictions of key parameters.
4. Component and/or breadboard validation in laboratory environment.	Following successful "proof-of-concept" work, basic technological elements must be integrated to establish that the pieces will work together to achieve concept-enabling levels of performance for a component and/or breadboard. This validation must be devised to support the concept that was formulated earlier and should also be consistent with the requirements of potential system applications. The validation is relatively "low-fidelity" compared to the eventual system: it could be composed of ad hoc discrete components in a laboratory.	A low fidelity system/component breadboard is built and operated to demonstrate basic functionality and critical test environments, and associated performance predictions are defined relative to the final operating environment.	Key, functionally critical, software components are integrated, and functionally validated, to establish interoperability and begin architecture development. Relevant Environments defined and performance in this environment predicted.	Documented test performance demonstrating agreement with analytical predictions. Documented definition of relevant environment.

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Appendix 7 - External Project References

Project Google Drive: https://drive.google.com/drive/folders/1YljWkiZftO_0Xt_Q3BoYnPZmuSWz85Dq?usp=drive_link

Project Timeline: https://docs.google.com/spreadsheets/d/1WLprxVUaz45zchfinad5Hi4OvU1nBjAnH-qq9N-64rA/edit?usp=drive_link

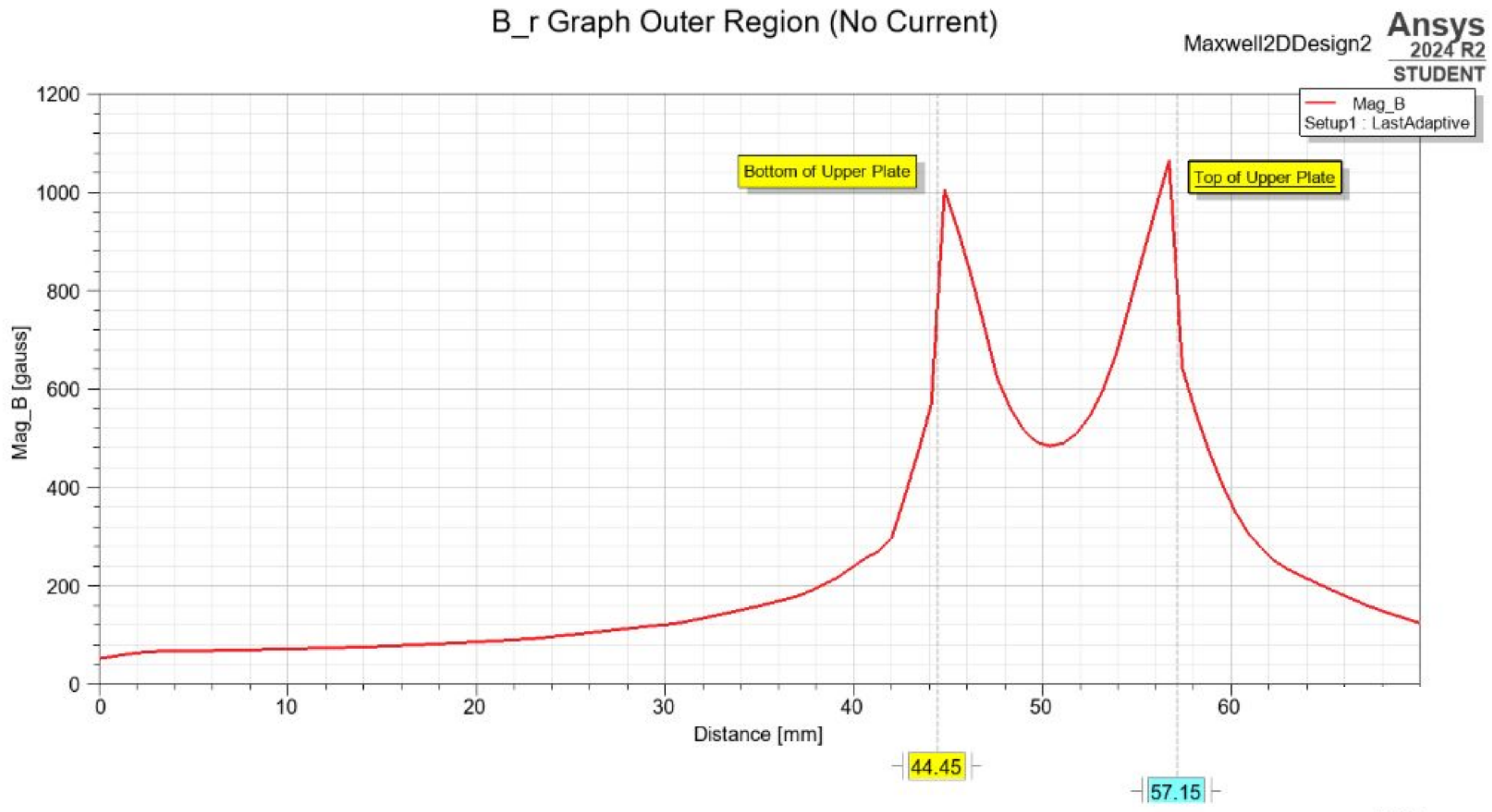
Project Budget: https://docs.google.com/spreadsheets/d/1OUD2Nij2bswQg2wwDJrrHRy2rcDOyuUAXW3beRVUSjQ/edit?usp=drive_link

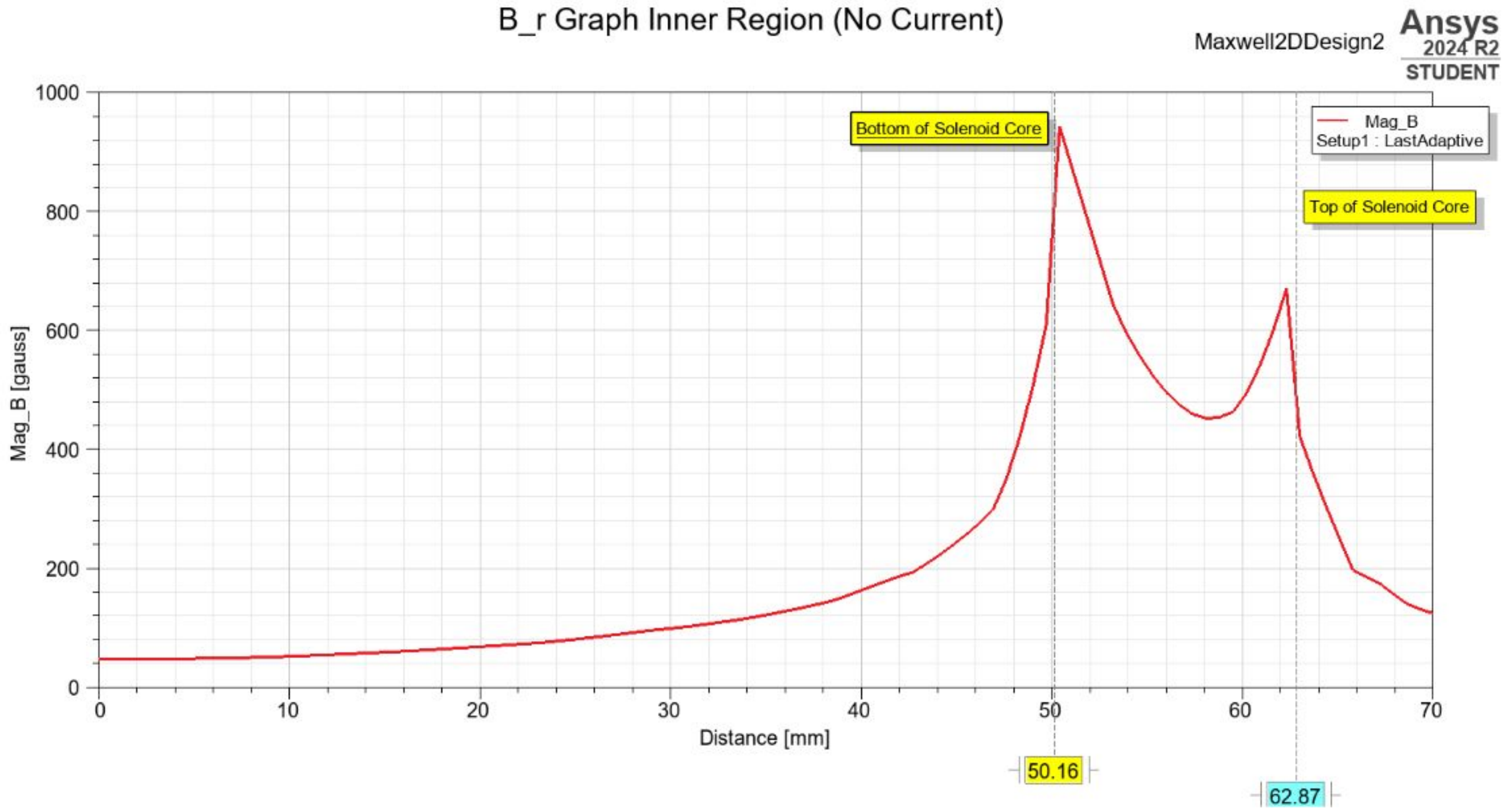
Standard Operating Procedures Document: https://docs.google.com/document/d/12Vz6qfsODYrluscqhkR-yc270HZ3h67F7TGFgSeCJVs/edit?usp=sharing

Real-time Electronic Monitoring program: https://drive.google.com/drive/folders/1Hc-604NB7RG46H_BqMUB5aAMip-PTkCl?usp=drive_link

Ansys Maxwell Simulation & Results Files: https://drive.google.com/drive/folders/1liKYHiHT8wF1NmpS4nHtqDTjzJbioLS0?usp=drive_link

Backup Slides





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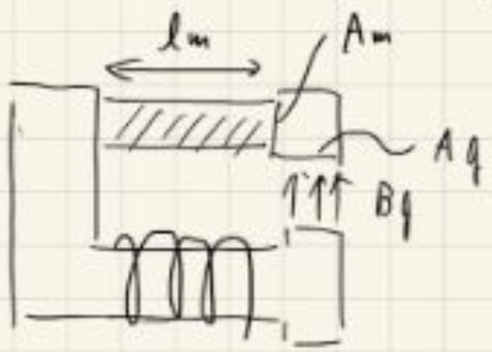

Continuity eq.

$$\nabla \cdot B = 0 \rightarrow B_m A_m = B_g A_g$$

$$\Leftrightarrow B_m = \left(\frac{A_g}{A_m}\right) B_g$$

$$\rightarrow \begin{cases} (B_m)_{exp} = \left(\frac{A_g}{A_m}\right)_{exp} B_{g\,exp} \\ (B_m)_{sim} = \left(\frac{A_g}{A_m}\right)_{sim} B_{g\,sim} \end{cases}$$

$$B_{g\,exp} = B_g$$

$$\rightarrow (B_m)_{sim} = \left(\frac{A_g}{A_m}\right)_{sim} \Big/ \left(\frac{A_g}{A_m}\right)_{exp} B_{m\,exp}$$

where $\left(\frac{A_g}{A_m}\right)_{sim} = \frac{15}{12.7}$

$$\left(\frac{A_g}{A_m}\right)_{exp} = \frac{15 \times 2\pi \times 54.61}{4 \times \pi \times (12.7/2)^2}$$

# of magnets

$$\rightarrow \left(\frac{A_g}{A_m}\right)_{sim} \Big/ \left(\frac{A_g}{A_m}\right)_{exp} \sim 0.1163$$

$$\rightarrow (B_m)_{sim} = 0.116\,(B_m)_{exp}$$

Area Ratio:

Simulated/Actual = 0.116